Who bears the AI shock?
The distributional and fiscal incidence in the UK*

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Abstract

I study how the household and fiscal costs of an AI labour-market shock in the United Kingdom depend on who bears it. I link occupation-level AI exposure to workers in the Family Resources Survey 2024–25 as prepared for PolicyEngine UK, an open-source tax–benefit microsimulation model, and pass the implied employment, wage and capital-income shocks through the model. I vary the shock along two axes: 66 shock-size combinations, and five incidence families that hold the aggregate shock fixed while varying who is displaced. In the central scenario—7 per cent of employees displaced—the annual Exchequer cost is £18.2 billion and before-housing-costs poverty rises by 1.75 percentage points. Reallocating the same aggregate shock moves the fiscal cost from £14.9 billion under uniform incidence to £22.1 billion under expertise compression, and £32.6 billion under an author-constructed, top-loaded stress test, while poverty changes far less. Income inequality rises in every scenario. The incidence of the shock determines whether it arrives as a fiscal problem or a poverty problem.

Keywords: artificial intelligence, labour market, microsimulation, poverty, inequality, public finances

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1 Introduction

If artificial intelligence displaces a substantial share of UK employment over the next decade, who pays—the displaced workers, the poorest households, or the Exchequer? The answer turns out to depend less on how large the shock is than on a question that policy analysis has so far left implicit: *who bears it*. The UK's services-intensive economy concentrates employment in the professional, financial, administrative and managerial occupations that score highest on standard measures of AI exposure (Felten et al., 2021; Eloundou et al., 2024; Pizzinelli et al., 2023), while its safety net—means-tested, tapered, and financed overwhelmingly by taxes on labour income—was built to insure shocks that strike the bottom of the income distribution. A displacement shock aimed at the top of the earnings distribution implicates the welfare state twice over: its insurance function faces claimants it rarely serves, and its revenue base sits precisely in the labour income the shock destroys.

Yet the top-loaded incidence implied by exposure indices is an assumption, not a fact. The empirical literature on AI's labour-market effects has not settled where displacement lands. Exposure-based measures place the risk on high-paid cognitive work (Felten et al., 2021; Eloundou et al., 2024); a competing view holds that AI compresses the returns to accumulated expertise, threatening the wage premia of experienced professionals while opening high-value work to less credentialed workers (Autor, 2024); and the earliest firm-level evidence finds job losses concentrated among *junior* workers within exposed occupations (Klein Teeselink, 2025; Hosseini and Lichtinger, 2026a; Brynjolfsson et al., 2025a). These are not refinements of a single estimate. They are different answers to the incidence question, and they imply different claimants at the benefit office and different holes in the tax base.

This paper makes incidence a first-class scenario axis. Rather than committing to one view of who bears the shock, it runs the competing incidence assumptions—not just competing shock sizes—through a complete tax and transfer system, turning a theoretical disagreement about who is displaced into a measured difference in poverty, inequality and Exchequer cost. Using PolicyEngine UK, an open-source microsimulation model running on its processed Family Resources Survey 2024–25 microdata, with AI exposure assigned to Standard Occupational Classification 2020 (SOC2020) occupations (Felten et al., 2021; Tolan et al., 2021; Eloundou et al., 2024), I simulate a common aggregate shock—seven per cent of employees displaced, a calibration convention within the range implied by Briggs and Kodnani (2023)—under four incidence families: displacement proportional to occupational exposure, displacement concentrated on junior workers, a stylised expertise-compression scenario in which senior premia bear the shock, and a uniform comparator. Around the central calibration I simulate a grid of 66 shock-size combinations crossing displacement rates with wage uplifts, with the capital channel applied throughout. Every result can be regenerated from public code; most comparable exercises rest on closed models and restricted microdata. I then take the framework to policy: a fifth incidence family calibrated to measured UK firm-level displacement patterns, three simulated fiscal responses in the shocked world, a tax-composition accounting of how far corporate profits could refill the revenue hole, and a mapping of the shock to the 650 Westminster constituencies.

Two findings organise the paper. First, one result is robust to everything I vary: the Gini coefficient of equivalised household disposable income rises in all 66 shock-size cells (by 0.06 to 1.55 percentage points) and in all five incidence families (by 0.93 to 1.13 percentage points at the central shock). Inequality rises regardless of who bears the shock—even though, in the exposure family, the market shock itself is progressive in the sense of falling hardest on the highest deciles. The tax-benefit system claws back part of high earners' losses through forgone tax rather than replacing them through benefits, so a top-loaded market shock still widens the disposable-income distribution.

Second, incidence determines the *form* the cost takes. The same 1.6 million displaced workers cost the Exchequer £18.2 billion when allocated by exposure (junior-concentrated incidence is statistically indistinguishable at £18.3 billion), £22.1 billion under expertise compression, and £14.9 billion under uniform incidence—a spread of £7 billion a year on an identical aggregate shock, all figures means over 20 displacement draws. Poverty moves less: before-housing-costs poverty rises by 1.75 to 1.93 percentage points across the four stylised families, a band within which no ordering survives the draw noise—while the measured family exceeds all four on both the fiscal and poverty margins (Section 4.5). Top-loaded incidence gives the UK a fiscal problem; flatter or junior-concentrated incidence is cheaper for the Treasury but leaves poverty about as high; compression delivers both problems at once. Which world the UK is in cannot yet be settled empirically—so I quantify what is at stake in the disagreement.

The incidence axis matters precisely because exposure indices cannot resolve it. Occupational exposure is age-blind: it cannot distinguish the junior analyst from the senior partner doing related work, so any scenario allocated by exposure alone spreads displacement across the age structure of exposed occupations, which in the UK peaks among prime-age workers. The junior-concentrated pattern in UK firm data (Klein Teeselink, 2025), US resume histories (Hosseini and Lichtinger, 2026a) and US payroll records (Brynjolfsson et al., 2025a) must therefore be imposed on the framework rather than derived from it, and the expertise-compression scenario is an author-designed stress test of the mechanism articulated by Autor (2024) rather than an estimate.

The remainder of the paper proceeds as follows. Section 2 situates the study in the live debate over AI's incidence and in the microsimulation literature on technology shocks and household income. Section 3 describes the exposure mapping to SOC2020 major groups, the displacement, wage and capital mechanics, the four incidence families, and the PolicyEngine UK framework. Section 4 presents the distributional incidence of the central shock, the scenario grid, the five incidence families including the measured family, the constituency geography, the age and gender dimensions, benefit caseloads and robustness checks. Section 5 turns to policy responses: reform counterfactuals in the shocked world and the tax-composition channel. Section 6 concludes, draws out the policy implications and sets out limitations and further work.

2 Related literature

My study sits at the intersection of two literatures: an unresolved debate about *where* in the labour market AI's displacement effects will land, and a microsimulation tradition that traces labour-market shocks through tax-benefit systems to household disposable income. I take the incidence debate seriously as a live disagreement—each of its schools becomes a scenario family in my design—and I extend the microsimulation tradition by making incidence itself the object of variation.

2.1 Measuring exposure to artificial intelligence

The modern exposure literature descends from the occupation-level automation probabilities of Frey and Osborne (2017), whose estimate that 47 per cent of US employment was at high risk of computerisation set the template of scoring occupations against machine capabilities. Subsequent work replaced whole-occupation probabilities with task- and ability-level measurement:

Webb (2019) matches patent text to job-task descriptions; Felten et al. (2021) link progress in ten AI application areas to the O*NET ability requirements of each occupation, producing the AI Occupational Exposure index I build on; and Tolan et al. (2021) connect AI benchmark results to cognitive abilities and tasks. The arrival of large language models prompted a further generation of measures targeted at generative AI specifically: the task-level rubric of Eloundou et al. (2024), the ILO's global assessment of jobs at risk of transformation rather than elimination (ILO, 2025), the UK government's occupational assessment (DSIT and DfE, 2023), and the GAISI index of Henseke et al. (2025). What no index can supply is incidence: an index records what AI *could* do to an occupation's task bundle, and translating it into who actually loses work requires assumptions the indices themselves cannot supply. Hosseini and Lichtinger (2026b) show that exposure is an incomplete statistic even for the supply side: their measure of the potential supply shift—the expansion of the qualified worker pool as generative AI lowers expertise requirements—varies widely across occupations with identical exposure scores, and they publish it as an alternative public occupation-level measure. The complementarity adjustment of Pizzinelli et al. (2023), which I implement in Section 3.1, is one such assumption; the incidence families of Section 3.4 treat the remainder of that translation as an explicit scenario choice.

2.2 The incidence debate

Exposure identifies which occupations AI can reach; it does not settle who within the labour market bears the adjustment. On that question the literature splits into four schools that disagree about both the direction and the site of the loss. I set out the mainstream reading of the exposure indices first, then the earliest ex-post evidence that has begun to test it, and finally the theories that dispute both.

The exposure school. The dominant approach takes the exposure indices at face value as a map of incidence: the occupations that score as most exposed are the ones expected to shed work, and, so read, the indices point to the top of the labour market. They consistently show exposure rising with education and earnings—professional, managerial and administrative occupations score highest, a pattern that recurs across European labour markets (Williamson et al., 2025) and, for the UK specifically, in the government's official assessment (DSIT and DfE, 2023) and in the GAISI index of Henseke et al. (2025). On this view AI breaks with the routine-biased pattern of earlier waves, in which computerisation hollowed out codifiable mid-skill work (Autor et al., 2003; Goos and Manning, 2007; Goos et al., 2014; Acemoglu and Restrepo, 2022), and instead aims at the top of the wage distribution. The IMF strand adds an important qualification: high exposure need not mean displacement, because the same workers may enjoy the largest complementarity gains. Cazzaniga et al. (2024) formalise this ambiguity, and Pizzinelli et al. (2023) document cross-country variation in the exposure–complementarity mix; Rockall et al. (2025) apply the framework to the UK and conclude that AI could *reduce* wage inequality if displacement falls disproportionately on high earners—unless complementarity among those same workers offsets it. Two studies have carried exposure-based scenarios through national tax-benefit models—for Ireland (Doorley et al., 2026) and, for earlier automation, across European countries (Doorley et al., 2023), the latter finding that progressive taxes and means-tested transfers muted the pass-through of rising wage dispersion into household inequality. Both take exposure-proportional incidence as given; neither varies it.

The junior-concentrated evidence. The earliest ex-post evidence on generative AI points to a gradient none of the theoretical schools quite predicted: within exposed occupations, adjustment is concentrated on *junior* workers. Klein Teeselink (2025) link generative-AI exposure to UK firm-level records and find employment reductions of around 4.5 per cent at more exposed firms, concentrated in junior positions; Hosseini and Lichtinger (2026a) document, in US resume and posting data, junior employment falling by around 9 per cent within six quarters of firm AI adoption while senior employment is unaffected; and Brynjolfsson et al. (2025a) find a 16 per cent relative employment decline among workers aged 22–25 in the most exposed US occupations since late 2022, with older workers largely spared. If incidence is graded by seniority as much as by skill, the distributional consequences differ from a classic skill-biased shock, because junior workers are spread across the household income distribution and interact differently with means-tested benefits.

The expertise-compression school. A competing view, associated with Autor (2024), holds that AI's distinctive property is not that it targets exposed occupations but that it compresses the returns to accumulated expertise. By supplying decision-relevant knowledge on demand, AI lets less experienced and less credentialed workers perform tasks previously reserved for elite professionals—an opportunity to rebuild middle-class work, but a threat to the wage premia of the experienced workers whose scarcity those premia reflect. Experimental evidence is consistent with the mechanism: generative AI raises output most for the least experienced workers, compressing within-occupation productivity gaps in customer support (Brynjolfsson et al., 2025b) and professional writing (Noy and Zhang, 2023). Hosseini and Lichtinger (2026b) formalise and quantify this mechanism: using LLM ratings of O*NET tasks they construct occupation-level potential supply shifts—the expansion of the qualified worker pool as generative AI lowers expertise requirements—and find in a calibrated general-equilibrium model that entry-barrier erosion compresses between-occupation wage inequality through wage adjustment rather than displacement, with losses concentrated in mid-to-upper-expertise scarcity premia and the extreme top largely insulated. On this view the shock lands on senior premia rather than junior heads—nearly the mirror image of the junior-concentrated evidence above, which is precisely why incidence needs to be varied rather than assumed.

Bounds on the shock. Two further strands constrain the disagreement rather than adding a scenario family of their own. The first bounds its *direction*: by making prediction cheap, AI raises the value of the human complements to prediction—judgement, decision-making and responsibility (Agrawal et al., 2018)—so that workers whose roles centre on decisions rather than predictable task execution may gain, pushing incidence towards prediction-like tasks and away from the senior decision-makers on whom the expertise-compression view lands it. The second bounds its *magnitude*: realised effects may simply be small. Danish population-scale evidence finds minimal effects of AI chatbots on earnings and hours in the first years of diffusion (Humlum and Vestergaard, 2025), US vacancy data show exposed establishments reorganising hiring without detectable aggregate employment effects (Acemoglu et al., 2022), and Acemoglu (2025) derives modest macroeconomic gains from task-level foundations. Acemoglu and Restrepo (2022) show that even historically transformative automation delivered only a cumulative 3.4 per cent TFP gain over 1980–2016 while reshaping the entire wage structure—large distributional effects do not require large aggregate ones. US evidence on publicly funded job training likewise shows displaced workers from AI-exposed occupations increasingly capturing large retraining returns by sorting toward less-exposed work (Hyman et al., 2025). Together they discipline both where the shock might land and how large a shock a simulation should entertain; my grid therefore extends down to small shocks.

2.3 From market incomes to disposable incomes

Whichever school proves right, the welfare consequences run through the tax-benefit system, and market-income incidence is a poor guide to disposable-income incidence. The European evidence on earlier automation makes the point directly: tax-benefit systems absorbed much of the routine-biased shock before it reached household inequality (Doorley et al., 2023). For AI, the factor-income channel compounds the case for full microsimulation: task-based models imply that automation lowers the labour share and channels income to capital owners (Acemoglu and Restrepo, 2018), Moll et al. (2022) show that automation raises returns to wealth itself, and Korinek and Stiglitz (2019) develop the theory of worker-replacing progress and the redistribution it calls for—with progressive capital taxation examined as a response by Bastani and Waldenström (2024). Because capital income is far more concentrated than labour income, this channel raises household inequality even where wage dispersion is unchanged, and only a model that jointly handles earnings, benefits, direct taxes and capital income can net the channels out.

2.4 Positioning

The exposure school has produced tax-benefit simulations; the other schools have produced theory and early ex-post estimates; to my knowledge no study has held a single aggregate shock fixed and passed each school's incidence gradient through the same tax-benefit model, so that the fiscal and distributional stakes of the disagreement can be read off directly. That is what I do. I implement each school's incidence gradient as an explicit scenario family—exposure-proportional, junior-concentrated, expertise-compression, and uniform—and let the UK tax and transfer system determine how much the disagreement matters for poverty, inequality and the public finances.

3 Data and methods

The analysis proceeds in three stages. I first attach an occupation-level measure of exposure to artificial intelligence to each worker in a representative UK household survey. I then translate an assumed economy-wide AI shock into person-level changes to employment, wages and capital income, under alternative assumptions about who bears the displacement. Finally, I pass baseline and shocked data through an open-source tax-benefit microsimulation model to recover the distributional and fiscal consequences. This section describes each stage; the incidence scenarios that form the paper's central axis are formalised in Section 3.4.

3.1 Measuring AI exposure

My starting point is the AI Occupational Exposure (AIOE) index of Felten et al. (2021), which links progress in ten AI application areas to the occupational ability requirements catalogued in O*NET, yielding a standardised score for each occupation that captures how much of its ability bundle overlaps with what AI systems can now do. Exposure alone does not distinguish occupations in which AI substitutes for workers from those in which it augments them.

I therefore combine the AIOE with the complementarity adjustment of Pizzinelli et al. (2023), who construct an index $\theta_l \in [0,1]$ from O*NET work-context descriptors — responsibility for others, physical proximity requirements, the consequence of error — and job-zone preparation requirements. High-θ occupations are those in which AI is likely to complement human judgement rather than displace it. The complementarity-adjusted exposure measure (C-AIOE) shields high-complementarity occupations from the raw exposure score:

$$\text{C-AIOE}_l = \text{AIOE}_l \times \big(1 - (\theta_l - \theta_{\min})\big), \tag{1}$$

where $\theta_{\min}$ is the minimum complementarity across occupational groups, so that the least complementary group retains its full exposure score. Alternative exposure measures include Tolan et al. (2021), the task-level rubric of Eloundou et al. (2024), and the cross-country applications of Cazzaniga et al. (2024); Section 4 reports the sensitivity of my headline results to substituting two of these for the C-AIOE.

Constructing the index for the UK requires a bespoke crosswalk, whose provenance I state in full because no published UK C-AIOE series exists at the level I need. AIOE scores are taken from the mapping to UK SOC2020 published with the DSIT analysis of AI and UK jobs (DSIT and DfE, 2023), itself derived from the Felten et al. (2021) scores via a chained crosswalk (US SOC2018 to US SOC2010, to ISCO-08, to UK SOC2020). The complementarity indices θ are not published at occupation level, so I reconstruct them from the public O*NET 27.3 release following the recipe documented by Pizzinelli et al. (2023); the reconstruction reproduces the published cross-occupation ordering up to a level offset, which is immaterial because equations (1) and (3) use θ only relative to its minimum and its mean. Both series are aggregated to the nine SOC2020 major groups using employment weights from the Annual Survey of Hours and Earnings (ASHE) 2025 Table 14, which covers 340 of the 412 four-digit unit groups; the remainder carry no ASHE employment estimate and are excluded from the weighting. Finally, the C-AIOE scores are shifted so that the least-exposed major group scores exactly zero. Under the allocation rule in Section 3.3 that group receives no job losses, and the shift prevents negative values of the standardised index from corrupting the allocation weights.

The one-digit level of aggregation is a genuine constraint. The Family Resources Survey (FRS) person records I use carry only the one-digit SOC2020 code, so exposure varies across just nine major groups and all within-group variation is lost. The exposure gradient I impose on the population is correspondingly coarse, and results should be read with that in mind; imputation of finer occupation codes from the Labour Force Survey is a planned refinement.

3.2 Shock scenarios

The size of the aggregate shock cannot be estimated from historical data and must be assumed; I treat it as a scenario input and vary it systematically. The central calibration takes a displacement rate of 7 per cent of employees and an aggregate wage uplift of 2.6 per cent for those who remain in work. Both figures are calibration conventions derived from Briggs and Kodnani (2023): the 7 per cent converts their task-exposure estimates into an employment displacement share, and the 2.6 per cent converts their productivity projections into an aggregate wage gain. Neither is a forecast, and I do not present them as one. Conceptually, the scenarios are calibrations of shocks to model primitives — occupation-level automation and productivity, as in the task framework of Acemoglu and Restrepo (2022) — whose labour-market consequences the displacement and wage rates then operationalise in reduced form, with the scenario grid below spanning the range of primitive calibrations in the literature. Alongside the labour-market effects, AI adoption is assumed to raise the return to capital by 0.4 percentage points, from a baseline of 1.005 per cent to 1.405 per cent. A permanent automation-induced rise in the return to wealth is exactly the mechanism formalised by Moll et al. (2022), whose measured increase in the return to investors' wealth of 0.5–2 percentage points since 1980 brackets the shock assumed here from above.

Because displacement and wage growth are genuinely uncertain, the central calibration sits inside a grid spanning displacement rates from 0 to 10 per cent and aggregate wage growth from 0 to 5 per cent — 66 scenario cells in all, with the capital-return shock applied throughout. The 1 per cent "low" anchor is a pure sensitivity case: Acemoglu (2025) estimates a ten-year GDP effect of around 1 per cent, not an employment displacement rate, and I do not attribute the anchor to him. Beyond the top of the grid, the 13 per cent "high" preset stress-tests a displacement rate roughly double the central convention.

3.3 Implementing the shock at the micro level

Employment shock. The aggregate number of displaced workers is the scenario displacement rate multiplied by weighted employee employment. The universe is all employees: workers whose SOC code cannot be matched to an exposure score form a pseudo-group assigned the employment-weighted mean exposure, so no employee is mechanically excluded from risk. The total is allocated across occupational groups l in proportion to each group's employment multiplied by its exposure:

$$\text{JobLoss}_l = \text{TotalJobLoss} \times \frac{\text{EMP}_l \times \text{C-AIOE}_l}{\sum_l \text{EMP}_l \times \text{C-AIOE}_l}, \tag{2}$$

so job losses concentrate in large, highly exposed groups and the zero-scored least-exposed group loses none. Within each group, displaced individuals are selected by ordering group members on independent uniform random keys and filling the group's weighted quota from equation (2), with the record straddling the quota boundary included with probability equal to the remaining shortfall over its weight. Because the ordering keys are uniform, a record's inclusion probability does not depend on its grossing weight; the weights enter only through the quota accounting, which is the design intent.

Displacement is modelled as full withdrawal from employee work. A displaced worker's employment income, hours, employee pension contributions and salary-sacrifice amounts are set to zero, any statutory pay is removed, and labour-market status is set to unemployed before the shocked simulation runs. Dual jobholders retain their self-employment income; only the employee job is lost. This is deliberately the starkest reading of displacement: in the task framework of Acemoglu and Restrepo (2022), task displacement "does not need to be associated with job loss" and manifests in equilibrium as relative wage declines and reallocation rather than unemployment. The full-withdrawal treatment should therefore be read as a short-run, within-year upper bound on an adjustment that a wage-margin model would spread over earnings; the duration sensitivity in Section 4.3 quantifies how quickly the bound decays. PolicyEngine UK takes no unemployment-duration input, so displaced workers are assessed as current-period unemployed under the model's standard benefit rules rather than as long-term unemployed with contributory entitlements exhausted. To the extent that new-style (contributory) Jobseeker's Allowance would in practice be exhausted within a year, my results may modestly overstate the benefit support reaching displaced workers; I flag this in Section 6.2.

Wage shock. Each worker who survives the employment shock receives a percentage uplift proportional to complementarity, normalised by the employment-weighted mean complementarity over baseline workers:

$$\text{WageChange}_l = \text{WageGrowth} \times \frac{\theta_l}{\bar{\theta}} \times \text{Wage}_l, \qquad \bar{\theta} = \frac{\sum_l \text{EMP}_l\, \theta_l}{\sum_l \text{EMP}_l}, \tag{3}$$

where EMP_l is baseline (pre-shock) employment. This is the baseline wage rule. Two properties should be noted. The normalisation makes the *employment-weighted mean* percentage uplift equal the assumed wage growth. The aggregate wage bill grows by exactly that rate only if θ is uncorrelated with pay. Since high-θ occupations earn more, the wage-bill-weighted uplift is somewhat larger. A second property: because $\bar{\theta}$ is computed over all baseline workers while the uplift accrues only to survivors—who, being drawn from less-exposed occupations, have above-average θ—displacement raises the realised mean survivor uplift slightly above the headline rate. Exact conservation is verified in seeded tests at zero displacement; both departures are properties of the rule itself rather than implementation error. The economic logic mirrors equation (1): AI complements rather than substitutes for high-θ occupations, so those occupations capture a larger share of the productivity gains — an assumption the compression scenario of Section 3.4 deliberately inverts. In the decomposition of Acemoglu and Restrepo (2022), this uplift plays the role of the economy-wide productivity effect, while the displacement effect is carried here entirely by the employment margin; their ripple effects — displaced workers bidding down the wages of the non-displaced — are outside the framework, so non-displaced workers can only gain.

Capital shock. The 0.4 percentage-point increase in the return to capital is implemented by scaling each person's savings-interest and dividend income by the ratio of shocked to baseline return, $1.405/1.005 \approx 1.398$. The baseline return of 1.005 per cent is taken from the baseline calibration; because the scaling factor is the ratio of two return levels, the size of the capital channel is highly sensitive to this assumption. The same +0.4 percentage-point shock applied to, say, a 4 per cent baseline return would scale capital incomes by only 10 per cent rather than 40 per cent. The capital-channel results should therefore be read as conditional on this calibration; equivalently, the shock can be read directly as a 39.8 per cent proportional increase in interest and dividend income. In the framework of Moll et al. (2022) the relevant return is that on investors' wealth rather than the safe rate, so a refinement would load the shock onto dividend income alone; because equity income is more concentrated than interest income, the uniform scaling here is, if anything, conservative about the top-end incidence of the capital channel. Rental income is excluded from the scaling, on the modelling assumption that the AI productivity channel operates through financial rather than housing capital. The self-employed are outside both labour-market shocks; their interest and dividend income is scaled like everyone else's.

3.4 Incidence scenarios

The rules above fix how large the shock is; they do not settle who bears it, and the empirical literature has not settled that question either. I therefore treat incidence as a first-class scenario axis. Holding the central aggregate shock fixed — 7 per cent displacement, a 2.6 per cent wage uplift, the capital shock — I re-allocate the same weighted quota of displacement under four incidence families:

Exposure-proportional. The default allocation of equations (2) and (3): displacement follows the C-AIOE gradient and wage gains follow complementarity. This family is the central calibration used throughout the shock-size grid.

Junior-concentrated. Displacement draws are tilted towards workers aged 16–24 by a multiplier equal to the ratio of junior to overall displacement effects reported by Klein Teeselink (2025) for the UK (5.8/4.5), consistent with the seniority-biased evidence of Hosseini and Lichtinger (2026a) and Brynjolfsson et al. (2025a). The group quotas and wage rule are otherwise unchanged.

Expertise-compression. A stylised, author-designed stress test of the view that AI compresses expert rents rather than displacing routine work. Displacement draws are tilted (multiplier 2) towards the top weighted-earnings tertile of workers in above-median-exposure occupations, and the wage uplift is allocated by *inverse* complementarity — θ is reflected about its range, $\theta'_l = \theta_{\max} + \theta_{\min} - \theta_l$, so that mid-skill workers capture the capability gains — with the same employment-weighted normalisation as equation (3). The formal treatment of this channel in Hosseini and Lichtinger (2026b) operates through wage compression for continuing workers rather than job destruction; the displacement-based implementation here should therefore be read as a fiscal stress test of the same incidence gradient, not as an implementation of their model. Their finding that wage gains accrue at the bottom of the expertise distribution while mid-to-upper scarcity premia erode also supports the inverse-complementarity wage rule used in this family.

Uniform. Every employee is assigned equal exposure and equal complementarity, so displacement risk and wage gains are flat across occupations. This family isolates the contribution of the incidence gradient itself: differences between it and the others are attributable entirely to who bears the shock.

Each family delivers the same weighted count of displaced workers to within rounding, so cross-family differences in fiscal cost, poverty and inequality measure the consequences of incidence alone. The incidence gradient in each family is an assumption, not an estimate; the paper's contribution is to show how the tax-benefit system mediates whichever incidence materialises.

Measured-incidence family. Section 4.5 adds a fifth family that replaces the stylised gradients above with the displacement patterns measured in UK firm-level data by Klein Teeselink (2025). Displacement probability is proportional to the shifted C-AIOE multiplied by three tilts: a junior multiplier of 1.29 (the ratio of their junior to overall displacement effects, 5.8/4.5), a London multiplier of 1.5, and a wage-tier multiplier of 9.6 above versus 0.6 below weighted-median earnings, reflecting their finding that losses concentrate in high-paying firms; the aggregate shock stays at the central preset. The reader should treat this family as an author-constructed, top-loaded stress test loosely anchored to Klein Teeselink (2025) rather than an implementation of it.[1]

[1] Verification against the full text (SSRN working paper, December 2025 revision) demands more candour than a citation conveys. The paper's own estimates are difference-in-differences semi-elasticities per standard deviation of LLM exposure: total employment −0.3 per cent, junior employment −0.4 per cent (a ratio close to the 1.29 used here, though built from two rounded coefficients of which the senior effect is statistically insignificant), and high-compensation firms −0.7 per cent against a statistical zero for low-compensation firms. The 9.6/0.6 wage-tier figures descend from the accompanying press release's scaling to maximally exposed firms, not from

3.5 Microsimulation with PolicyEngine UK

I use PolicyEngine UK (version 2.89.2), an open-source tax-benefit microsimulation model, running on PolicyEngine UK's processed (base, unenhanced) Family Resources Survey 2024–25 microdataset. This reshapes the UK Data Service FRS 2024–25 (study SN 9563) into PolicyEngine's tax-benefit schema, applies source-level imputations (council tax, benefit take-up, salary sacrifice) and retains the survey's own grossing weights; I obtain it from PolicyEngine's `policyengine-uk-data` repository and load it through `UKSingleYearDataset` and `Microsimulation`. Occupation codes are not carried in the processed dataset, so I join the SOC2020 major group from the raw FRS `adult.tab` file (UK Data Service SN 9563) onto the simulation's person table using the identifier convention person_id = SERNUM × 1000 + PERSON, which matches every FRS adult. The simulation year is 2026.

The counterfactual design is a paired comparison on identical data: a baseline simulation and a shocked simulation are run on the same records, with the shocked run receiving the modified employment income, hours, pension contributions, savings-interest income, dividend income and employment status of Section 3.3 as direct inputs. All reported effects are differences between the two runs. This design removes between-run noise — the survey sample and grossing weights are identical on both sides, so no part of a reported effect reflects resampling — but it does not quantify population sampling uncertainty: both runs inherit whatever sampling error the FRS itself carries, and my Monte Carlo intervals speak only to the stochastic identity of the displaced, not to survey sampling.

All income results are expressed on the HBAI cash disposable income concept — equivalised household net income as defined for the Households Below Average Income statistics and implemented in PolicyEngine UK — which is also the concept underlying the poverty lines. I report four outcome families: the change in the government balance (the Exchequer cost of the shock, net of additional revenue on higher wages and capital income); person-weighted poverty rates before and after housing costs (BHC and AHC); the Gini coefficient of equivalised household disposable income, computed with negative incomes bottom-coded at zero; and mean changes in household income by decile, with deciles fixed at their baseline positions so that individuals are tracked rather than re-ranked.

Because the identity of the displaced is stochastic, I adopt explicit seed conventions and label every result accordingly. The 66-cell grid, the three presets and the decomposition figure are single draws at seed 0, so that cells differ only in their scenario inputs, never in their random draw. The five incidence families report means over 20 draws (seeds 0–19). The decile transition and wage profiles (Figures 1 and 2) average 50 seeded draws. The robustness summary and the decomposition confidence band use a 20-draw Monte Carlo at seeds 0–19; wherever a ± value appears it is the standard deviation across draws, not a survey-based standard error.

Policy reforms in the shocked world. The reforms of Section 5 are evaluated as $M(\text{shock}+\text{reform}) - M(\text{shock})$ on the fixed seed-0 shock draw, applying for 2026 only, with poverty lines held at their shocked-world values. Benefit-side reforms are implemented as PolicyEngine UK

the paper's tables, and the pay split is by *firm average compensation*, which I proxy by person earnings. Most seriously, the London tilt of 1.5 is *contradicted* by the revised full text, in which London-headquartered firms show minimal, statistically insignificant employment effects while the losses concentrate outside London; I retain the tilt so that the family matches its originally circulated form. Section 4.5 shows the family's fiscal and poverty results are robust to replacing 9.6/0.6 with much milder ratios; the multiplicative independence of the three channels and the age-under-25 proxy for their seniority concept remain author-imposed throughout.

parameter reforms — scaling the Universal Credit (UC) standard allowance, raising the benefit-cap thresholds and cutting the taper rate — so their costs are net of the means-testing clawback the model computes. Wage insurance is instead a direct post-simulation transfer to displaced workers (half of lost earnings, capped), treated as non-taxable and disregarded by means tests, so its gross cost equals its net cost and brackets the generous end of possible implementations.

Constituency geography. The constituency results of Section 4.6 rest on PolicyEngine's $650 \times 53{,}508$ weight matrix, which assigns each household in the *enhanced* FRS 2023–24 dataset a calibrated weight in every Westminster parliamentary constituency. The SERNUM-based occupation join of Section 3.5 is invalid on that dataset, so SOC major group is imputed from the plain-FRS 2024–25 weighted distribution within age band × gender × region × earnings-decile cells (99.4 per cent weighted coverage from the full four-way cell, seed 0). Constituency variation therefore reflects demographic and earnings composition plus a single displacement draw, not observed occupation mix, and the geographic results — computed on a different dataset and period — are not directly comparable to the poverty numbers elsewhere in the paper.

4 Results

I build the results outward from the central scenario. Sections 4.1–4.3 trace that scenario from the three market-income channels through household mediation to the fiscal and distributional aggregates, and Section 4.4 widens the lens to a 66-cell grid over shock sizes. Section 4.5 then turns to the paper's central question—who bears the shock—by holding the aggregate shock fixed and varying its incidence across five families, four stylised and one shaped by measured UK evidence, before Sections 4.6–4.8 map the shock across parliamentary constituencies, the age and gender dimensions, and benefit caseloads. Throughout, all income and inequality results are computed on the HBAI cash disposable income concept described in Section 3.5. Seed and draw conventions follow Section 3.5 and Appendix A.1: the grid, the decomposition and the preset table are single draws at seed 0, the decile profiles average 50 draws, and the incidence-family and reform tables report means over 20 Monte Carlo draws; ± values are standard deviations across those 20 draws.

4.1 The three market-income channels

The central scenario displaces 7 per cent of employees—1.61 million workers, allocated across occupations in proportion to exposure—and grants surviving workers in exposed occupations a wage uplift averaging 2.6 per cent, alongside the capital income channel. Figure 1 plots the share of each equivalised disposable income decile that transitions from employment to unemployment. The gradient is strictly monotone: 0.50 per cent of the bottom decile transitions, rising through every decile to 5.05 per cent at the top. It is important to be clear about what this figure shows. The gradient is not an empirical finding about who AI will in fact displace; it is the exposure-proportional allocation assumption made visible. Because the professional, associate-professional and administrative occupations that score highest on the C-AIOE index sit in the upper half of the income distribution, allocating displacement in proportion to occupational exposure necessarily concentrates job loss among higher-income households. The independent Irish estimates of Doorley et al. (2026), built on different microdata and a finer occupational classification, display a similarly top-concentrated gradient, which suggests the pattern is a general property of services-intensive economies under exposure-proportional incidence—but the incidence itself remains an assumption, and Section 4.5 treats it as one.

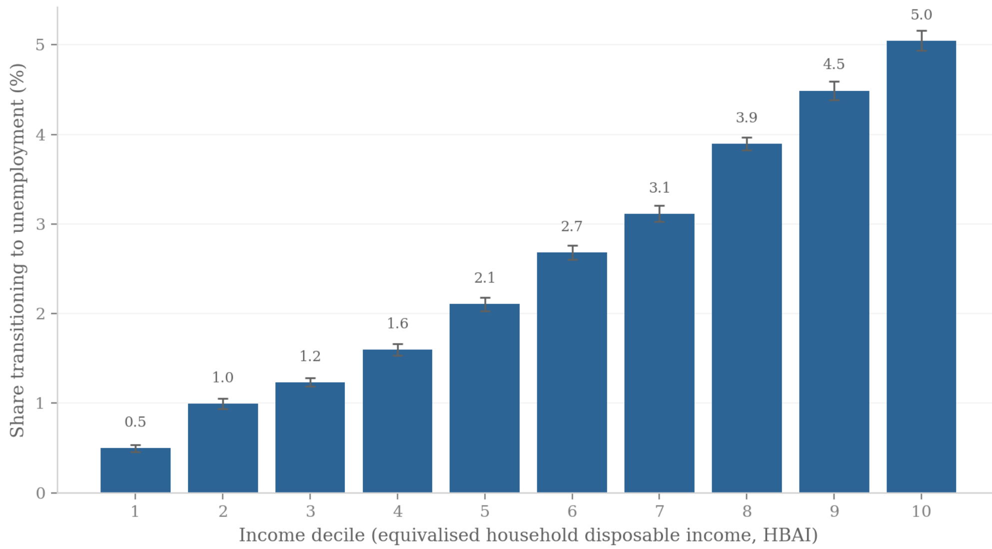

Figure 1: Share of the population transitioning from employment to unemployment by equivalised disposable income decile, central scenario. Bars are means over 50 seeded displacement draws; error bars show 95 per cent confidence intervals across draws. The gradient reflects the exposure-proportional allocation assumption, not observed incidence (see text).

The wage channel, by contrast, is almost distributionally flat. Figure 2 shows the percentage change in employment income among workers who remain employed: gains run from 2.51 per cent in the bottom decile to 2.72 per cent in the top, a spread of barely 0.2 percentage points around the 2.6 per cent average uplift. Exposure is sufficiently widespread across the deciles in which people work that the productivity dividend is spread thinly and evenly. It is the displacement margin, not the wage margin, that drives the distributional results below. That allocation is a property of the scenario design rather than a finding: in the US evidence of Acemoglu and Restrepo (2022), task displacement operates chiefly through relative wage declines, which account for 50–70 per cent of changes in the wage structure. A wage-margin implementation of the same aggregate shock would shift losses from benefit caseloads to earnings, with smaller fiscal costs and a different poverty profile.

The third channel is capital income. The scenario assumes part of the productivity gain is capitalised into returns on existing capital, applying a uniform proportional uplift to interest and dividend income (Section 3.3). Although the percentage change is identical across deciles by construction, Figure 3 shows the incidence in pounds is heavily concentrated: the top decile holds 35.2 per cent of all capital income (£8.1 billion of the baseline aggregate in my data), the top two deciles together more than half, and the bottom decile just 2.8 per cent. Whatever share of AI gains is capitalised flows overwhelmingly to households already at the top of the distribution.

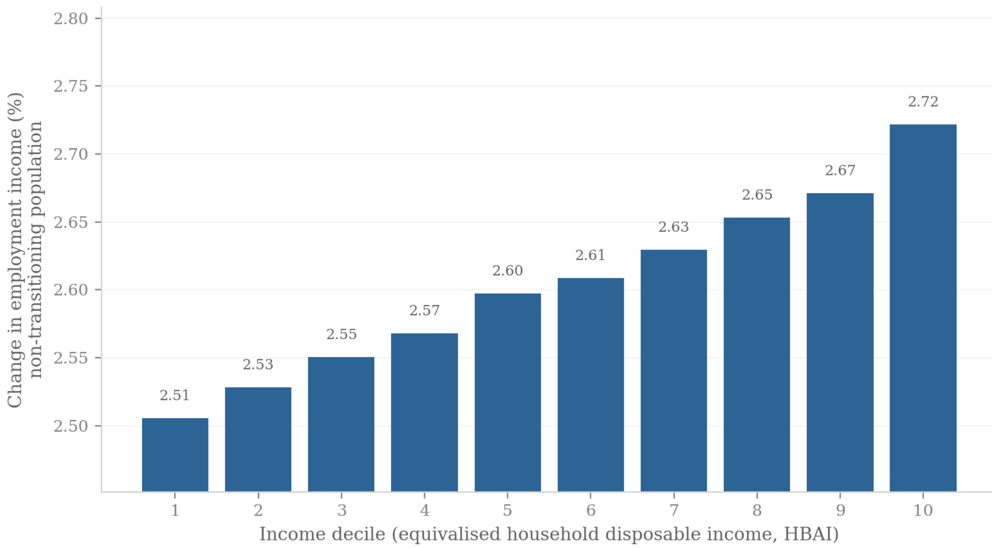

Figure 2: Percentage change in employment income among workers remaining in employment, by income decile, central scenario. Means over 50 seeded draws.

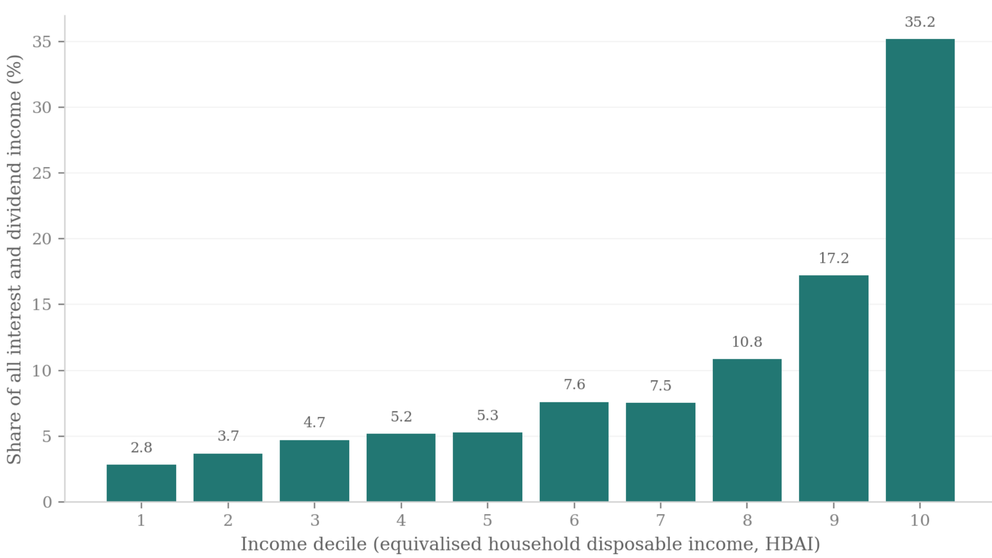

Figure 3: Capital income channel by decile: a uniform proportional uplift, but pound gains concentrated where capital income is held.

4.2 Household mediation of the shock

The tax-benefit system converts each market-income loss into a smaller disposable-income loss; Figure 4 decomposes that mediation by decile into market income, benefits, and taxes and contributions, all expressed relative to baseline disposable income. The decomposition is household-level and additive: market income change plus benefit change minus tax change equals the disposable income change up to an explicit residual, which collects perimeter items inside the HBAI concept (council tax and pension deductions) and never exceeds 0.5 per cent of baseline income in any decile.

Market income losses follow the transition gradient: they are near zero in the bottom decile, around 1–1.6 per cent of baseline disposable income in deciles 3–5, and reach 6.2 and 5.3 per cent in deciles 9 and 10. The tax-benefit system then cushions these losses through two distinct mechanisms. Across the middle deciles, automatic benefit entitlements—Universal Credit in particular—rise by up to 0.5 per cent of baseline income as displaced earners qualify for means-tested support. At the top the cushioning operates through the tax system instead: deciles 9 and 10 see their income tax and National Insurance liabilities fall by 1.3–1.4 per cent of baseline income as high-marginal-rate earnings disappear. The combined offset is substantial. In decile 9, a market income loss of 6.2 per cent of baseline disposable income becomes a disposable income loss of 4.3 per cent—roughly three-tenths of the market shock absorbed—and in decile 5 a 1.6 per cent market loss becomes a 1.0 per cent disposable loss. Averaged over 20 displacement draws, disposable income losses run from 0.53 per cent in the bottom decile to 4.09 per cent in the top; the gradient is broadly upward and stable across draws, with only a minor decile-2/decile-3 inversion.

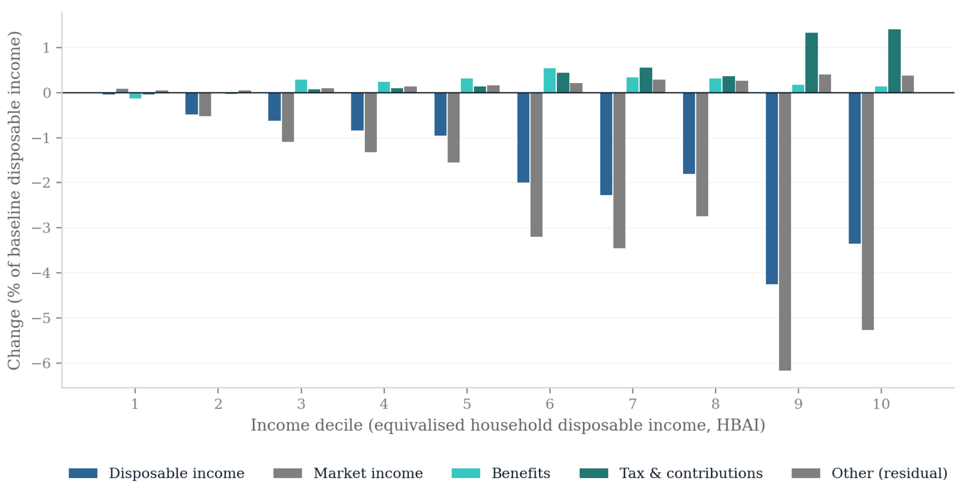

Figure 4: Additive household-level decomposition of the change in income by decile into market income, benefits, and taxes and contributions, central scenario (single draw, seed 0), as a share of baseline HBAI disposable income. The residual collects perimeter items (council tax, pension deductions) inside the HBAI concept.

4.3 Fiscal and distributional aggregates

Table 1 summarises the preset scenarios. In the central scenario the annual Exchequer cost is £18.4 billion, driven by lost income tax and National Insurance on displaced higher earners together with higher benefit spending. Poverty rises by 1.87 percentage points before housing costs (1.88 points after housing costs) and the Gini coefficient of equivalised disposable income rises by 1.10 percentage points from a baseline of 0.309. The low scenario—a 1 per cent displacement rate with no wage gain, which I treat purely as a sensitivity case rather than an estimate drawn from the literature—costs £2.5 billion and moves poverty and the Gini by less than 0.2 points each. The high scenario, at 13 per cent displacement, costs £46.7 billion a year with poverty up 3.80 points. PolicyEngine's before-housing-costs poverty line is an *absolute* threshold, so these poverty changes measure absolute impoverishment rather than movement relative to a shock-shifted median. On the relative (60 per cent of contemporaneous median) BHC line the central scenario raises poverty by $+0.85 \pm 0.11$ percentage points over 20 draws — about half the absolute-line rise, because the shock lowers the median itself and so drags the relative line down with it.

The scenario anchors deserve a note of caution. The central calibration converts the task-exposure and productivity figures of Briggs and Kodnani (2023) into a displacement rate and a wage uplift via the conventions described in Section 3.2; it is a calibration convention, not a forecast. The low scenario is likewise not an employment estimate from Acemoglu (2025), whose 0.9–1.1 per cent figure refers to ten-year GDP growth, not job displacement.

Table 1: Summary of preset scenarios (single draw, seed 0)

Scenario	Displaced (million)	Exchequer (£bn/yr)	Δ Poverty (BHC, pp)	Δ Gini (pp)
Low	0.23	2.5	+0.14	+0.19
Central	1.61	18.4	+1.87	+1.10
High	3.00	46.7	+3.80	+1.83

Note: Low: 1 per cent displacement, no wage gain (sensitivity case). Central: 7 per cent displacement, 2.6 per cent wage gain. High: 13 per cent displacement, 2.6 per cent wage gain. The capital-return shock (+0.4pp) applies in every preset.

None of this depends on the particular displacement draw. Repeating the central scenario over 20 independent draws gives a mean Exchequer cost of £18.2 billion (standard deviation £1.5 billion, range £15.8–21.1 billion), a poverty increase of $+1.75 \pm 0.11$ percentage points and a Gini increase of $+1.03 \pm 0.09$ points. Nor does it depend on the exposure index: rerunning the central scenario with the DSIT AIOE mapping or the GPT-exposure measure of Eloundou et al. (2024) in place of the C-AIOE moves the Exchequer cost by less than £0.2 billion and leaves the transition gradient essentially unchanged (top-decile transition shares of 4.9 and 4.7 per cent against 4.7 under the C-AIOE; bottom-decile shares of 0.4–0.5 per cent under all three).

Two assumptions do move the headline numbers, and I state them as first-order rather than burying them among caveats. The first is displacement duration. The central scenario removes a full year of earnings from every displaced worker; if displaced workers instead find work after six months (implemented in-model by halving their annual earnings), the Exchequer cost falls from £18.2 billion to £4.2 ± 0.9 billion and the BHC poverty rise all but vanishes ($+0.09 \pm 0.08$ points), over 20 draws. The collapse is more than proportional—a naïve halving of the full-year result would give £9.1 billion—because workers who keep half a year's earnings remain taxpayers and largely stay clear of means-tested support. Every fiscal and poverty magnitude in this paper therefore scales strongly with the expected out-of-work duration, and the full-year figures should be read as a within-year worst case. The second is benefit take-up. Imposing a 70 per cent Universal Credit take-up probability (drawn identically in baseline and shocked worlds) moves the cost to £19.0 ± 1.5 billion and the poverty rise to $+1.69 \pm 0.11$ points: the headline results are robust to take-up, which shifts cost between the Exchequer and household margins without changing the picture.

4.4 The shock-size grid

To move beyond the presets I simulate a grid of 66 scenarios, crossing eleven displacement rates (0 to 10 per cent of employees, allocated in proportion to exposure, as in the presets) with six wage uplifts (0 to 5 per cent), each combined with the capital income channel, all under exposure-proportional incidence. Figure 5 maps the change in average household disposable income. The corners bracket the range: with a 5 per cent wage gain and no job loss, average disposable income rises by 3.0 per cent; with 10 per cent displacement and no wage gain it falls by 5.2 per cent. The near-zero contour runs close to the grid diagonal—roughly one percentage point of wage growth offsets one percentage point of displacement in aggregate disposable income terms—so whether AI raises or lowers average living standards depends almost entirely on the balance struck between these two forces, about which the empirical literature remains genuinely uncertain.

Wage \ Displacement	0	1	2	3	4	5	6	7	8	9	10
5	3.0	2.4	1.8	1.2	0.6	0.0	-0.7	-1.2	-1.8	-2.4	-3.0
4	2.5	2.0	1.3	0.7	0.1	-0.5	-1.1	-1.7	-2.2	-2.8	-3.5
3	2.0	1.5	0.9	0.2	-0.3	-0.9	-1.6	-2.1	-2.7	-3.3	-3.9
2	1.6	1.0	0.4	-0.2	-0.8	-1.4	-2.0	-2.5	-3.1	-3.7	-4.3
1	1.1	0.5	-0.1	-0.7	-1.3	-1.8	-2.5	-3.0	-3.5	-4.1	-4.7
0	0.6	0.0	-0.6	-1.2	-1.7	-2.3	-2.9	-3.4	-4.0	-4.6	-5.2

Figure 5: Change in average household disposable income (per cent) across the 66-cell grid of displacement rates and wage uplifts (single draw, seed 0).

Figure 6 shows the corresponding fiscal picture. One measurement issue deserves note: in PolicyEngine's UK aggregates, income tax plus National Insurance minus total benefit expenditure (which includes the state pension) is a negative number, so expressing revenue changes relative to that net position produces uninformative percentages. I therefore express the net revenue change relative to gross income tax and National Insurance receipts. Two consequences follow. First, the grid's revenue measure is income tax plus National Insurance minus cash benefit expenditure, a narrower concept than the full government-balance measure behind the £18.4

billion headline cost of Section 4.3, which also includes indirect taxes and other levies; the grid's cash-tax cost of the central 7 per cent/2.6 per cent scenario is roughly £12 billion, and the two figures should not be read off the same surface. Second, on that narrower basis the grid spans a gain of £22.3 billion, or 7.2 per cent of receipts (5 per cent wage growth, no displacement), to a loss of £31.2 billion, or 10.0 per cent of receipts, in the worst cell (10 per cent displacement, no wage growth). At the central displacement rate, 7 per cent with no wage offset costs £21.5 billion (6.9 per cent of receipts), which a 3 per cent wage uplift roughly halves to £10.2 billion. As with disposable income, the break-even locus lies close to the diagonal.

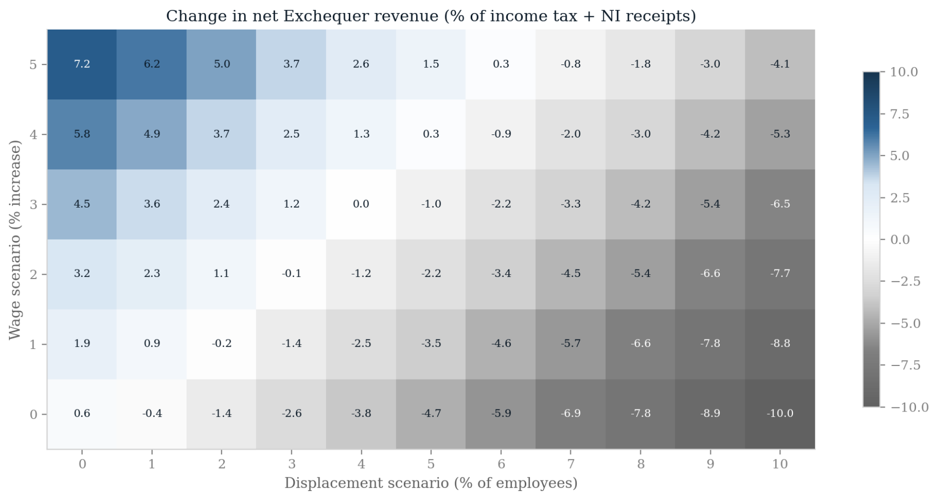

Figure 6: Net Exchequer revenue change as a percentage of gross income tax and National Insurance receipts, across the 66-cell scenario grid.

Figure 7 maps the change in the Gini coefficient across the same grid, and delivers the paper's most robust result: the Gini rises in all 66 cells, from +0.06 percentage points in the capital-only cell (no displacement, no wage gain) to +1.55 points at the far corner. No combination of parameters within the grid reduces measured inequality. The mechanism is one of polarisation rather than a simple rich-get-richer story, and it coexists with an incidence that is itself progressive: displacement strikes hardest at the top of the earnings ladder and pushes affected households down the distribution, while the wage uplift lifts the surviving employed—disproportionately in the upper-middle and top deciles—further away from workless and low-income households, and the capital channel reinforces the very top. Because both the loss side and the gain side of the shock widen gaps between households, inequality rises even though the market shock lands mainly on higher earners. That tension—a top-loaded shock that still raises the Gini—recurs throughout what follows.

4.5 The incidence axis: who bears the shock

Everything so far conditions on exposure-proportional incidence. But the allocation of a given aggregate shock across workers is at least as uncertain as its size, and the empirical literature points in different directions: exposure indices imply top-loaded incidence, the hiring-margin evidence points to juniors (Brynjolfsson et al., 2025a; Hosseini and Lichtinger, 2026a), and the expertise-compression view suggests AI erodes the premium of mid-skill judgement work. I

Figure 7: Change in the Gini coefficient of equivalised HBAI disposable income (percentage points) across the 66-cell scenario grid. The Gini rises in every cell.

therefore treat incidence as a first-class scenario axis. Holding the aggregate shock fixed at the central calibration—the same 1.6 million displaced workers, the same wage and capital channels—I reallocate displacement across five families. Four are stylised: *exposure-proportional* (the central calibration used above); *junior-concentrated*, which tilts displacement towards early-career workers; *expertise-compression*, an author-designed stress test in the spirit of the task literature descending from Autor et al. (2003), which concentrates displacement on mid-to-upper-skill occupations whose comparative advantage lies in codified expertise; and *uniform*, which allocates displacement occupation-neutrally. The fifth, *measured*, shapes displacement probabilities by tilts loosely anchored to the UK firm-level evidence of Klein Teeselink (2025) (with the provenance caveats of Section 3.4): exposure-proportional draws multiplied by a junior tilt (1.29), a London tilt (1.5) and a wage-tier tilt that loads displacement onto workers above median earnings (9.6 above, 0.6 below), while the aggregate shock stays at the central preset. Table 2 and Figure 8 report the results; the table gives means ± standard deviations over 20 Monte Carlo draws.

Three findings stand out. I call a ranking between two families *robust* only when their 20-draw means differ by more than one standard deviation, and treat them as indistinguishable otherwise; I adopt this convention once rather than repeat it at every comparison. First, who bears the shock determines what it costs. The Exchequer cost of an identical aggregate displacement ranges from £14.9 billion under uniform incidence to £32.6 billion under the measured family—a spread of nearly £18 billion a year, roughly the central estimate itself, generated purely by reallocating which workers lose their jobs. This fiscal ordering—measured above compression above the exposure/junior pair above uniform—is the only robust ranking in the table. Top-loaded incidence is fiscally expensive because displaced high earners take large income tax and National Insurance payments with them; uniform incidence is cheaper because uniformly drawn displaced workers earn, and therefore pay, less.

Second, the fiscal saving from flatter incidence buys no poverty relief. Across the four stylised families the BHC poverty rise sits in a narrow band of +1.75 to +1.93 points—the families are not reliably ordered—while their Exchequer costs span £7 billion. The uniform shock costs £3.3

Table 2: The same aggregate shock under five incidence families (central calibration; mean ± SD over 20 Monte Carlo draws; AHC column single draw, seed 0)

Incidence family	Displaced (million)	Exchequer cost (£bn/yr)	Δ Poverty BHC (pp)	Δ Poverty AHC (pp)	Δ Gini (pp)
Exposure-proportional	1.61	18.2 ± 1.5	+1.75 ± 0.11	+1.88	+1.03 ± 0.09
Junior-concentrated	1.61	18.3 ± 1.3	+1.77 ± 0.15	+1.83	+1.02 ± 0.08
Expertise-compression	1.61	22.1 ± 1.8	+1.93 ± 0.16	+2.08	+1.02 ± 0.12
Uniform	1.62	14.9 ± 1.9	+1.84 ± 0.13	+1.86	+1.13 ± 0.09
Measured (Klein Teeselink)	1.61	32.6 ± 2.0	+2.15 ± 0.14	+2.61	+0.93 ± 0.13

Note: The measured family combines the exposure-proportional draw with a junior multiplier (1.29), a London multiplier (1.5) and wage-tier multipliers (9.6 above / 0.6 below median earnings). Verification against the SSRN full text (December 2025 revision) shows the wage-tier figures descend from press-release scalings rather than the paper's per-SD estimates, and the London tilt is contradicted by the revised text; see Section 3.4. The family is an author-constructed top-loaded stress test loosely anchored to Klein Teeselink (2025). Its fiscal cost is robust to milder wage-tier ratios: £28.4 ± 2.0 billion under 3/0.8 and £31.3 ± 1.8 billion under 6/0.7, with poverty and Gini changes statistically indistinguishable across ratios (20 draws each).

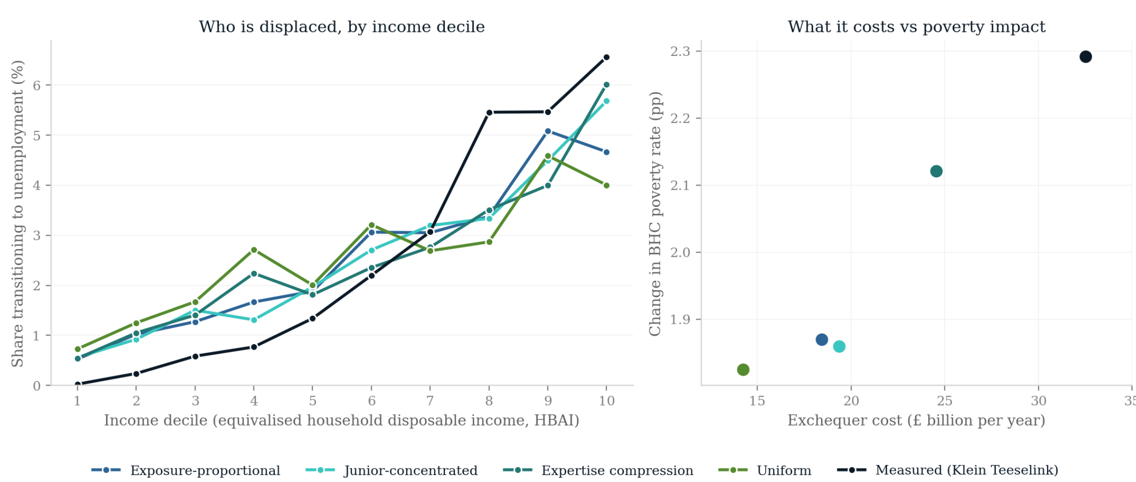

Figure 8: Decile incidence of the same aggregate central shock under five incidence families: share of each decile transitioning to unemployment, and change in equivalised HBAI disposable income by decile. The figure shows a single illustrative draw (seed 0), whereas the aggregates in Table 2 are means over 20 Monte Carlo draws.

billion less than the exposure-proportional one on the 20-draw means, yet its poverty rise is, if anything, slightly larger (+1.84 against +1.75 points). Cheap incidence is not benign incidence: the fiscal and poverty margins move separately, and a government cannot infer from a modest revenue hit that households near the poverty line have been spared.

Third, the top-loaded families are the worst of both worlds. Expertise compression costs £22.1 billion and raises BHC poverty by +1.93 points. The measured family goes further on both margins: at £32.6 billion it is robustly the costliest of the five, and its BHC poverty increase (+2.15 points; +2.61 AHC in the seed-0 draw) is robustly above the exposure, junior and uniform families (though not compression). Its wage-tier multiplier concentrates displacement on workers above median earnings—the most top-tilted decile gradient of the five families, with 0.02 per cent of the bottom decile transitioning against 6.6 per cent of the top—so it maximises the loss of income tax and National Insurance, while the affected households' fall from the middle and upper-middle of the distribution drags more households across the (absolute) poverty line than a purely top-decile shock would. To the extent that the measured UK evidence is a better guide to incidence than occupational exposure alone, my central preset understates both the fiscal and the poverty cost of the shock.

The Gini, meanwhile, rises in all five families—20-draw means of +0.93 to +1.13 percentage points—just as it rises in all 66 grid cells. No pairwise Gini ordering among the five families is robust, so I do not interpret which family is most unequalising; the robust statement is directional. Incidence reallocates the cost between the fiscal and poverty margins; it does not change the direction of the inequality effect.

4.6 The geography of the shock

The incidence question has a spatial counterpart: an occupationally selective shock lands unevenly across places because occupations cluster geographically. To map it I project the central scenario onto the 650 Westminster parliamentary constituencies using PolicyEngine's calibrated constituency weight matrix. One methodological caveat governs everything in this subsection. The weight matrix indexes the *enhanced* FRS 2023–24 dataset, on which occupation codes cannot be joined directly, so SOC major group is imputed from the plain-FRS distribution within age-band × gender × region × earnings-decile cells (99.4 per cent weighted coverage from the full four-way cell). Constituency variation therefore reflects demographic and earnings composition under a single seed-0 draw, not observed local occupation mixes, and the results are not directly comparable to the poverty numbers elsewhere in the paper: on this dataset and period the national BHC poverty response is roughly twice that of the central run, for the reasons decomposed in Appendix A.8. The enhanced dataset's calibrated capital incomes are moreover so much larger that the proportional capital shock is not fiscally comparable across datasets, which is why no Exchequer figures are reported in this subsection.

With that understood, the geography is far from uniform. Nationally, aggregate household income falls by 1.03 per cent and 71.5 workers per 1,000 are displaced. Across the twelve regions the income hit is deepest in Northern Ireland (−2.29 per cent) and Scotland (−1.79 per cent) and mildest in Wales (−0.78 per cent) and the South East (−0.83 per cent), even though displacement rates span only 61 (Northern Ireland) to 79 (Wales) per 1,000 workers: where displacement is similar, the income consequences depend on which earners are hit and how much cushioning the local benefit mix provides. At constituency level the spread widens further, combining Northern Irish seats (led by Lagan Valley, −5.43 per cent) with high-exposure inner-London ones, while a small number of constituencies gain on net, led by Kingston and Surbiton (+2.05 per cent), where wage and capital gains outweigh displacement losses; the full ranking is in Appendix A.8. Figure 9 maps both surfaces.

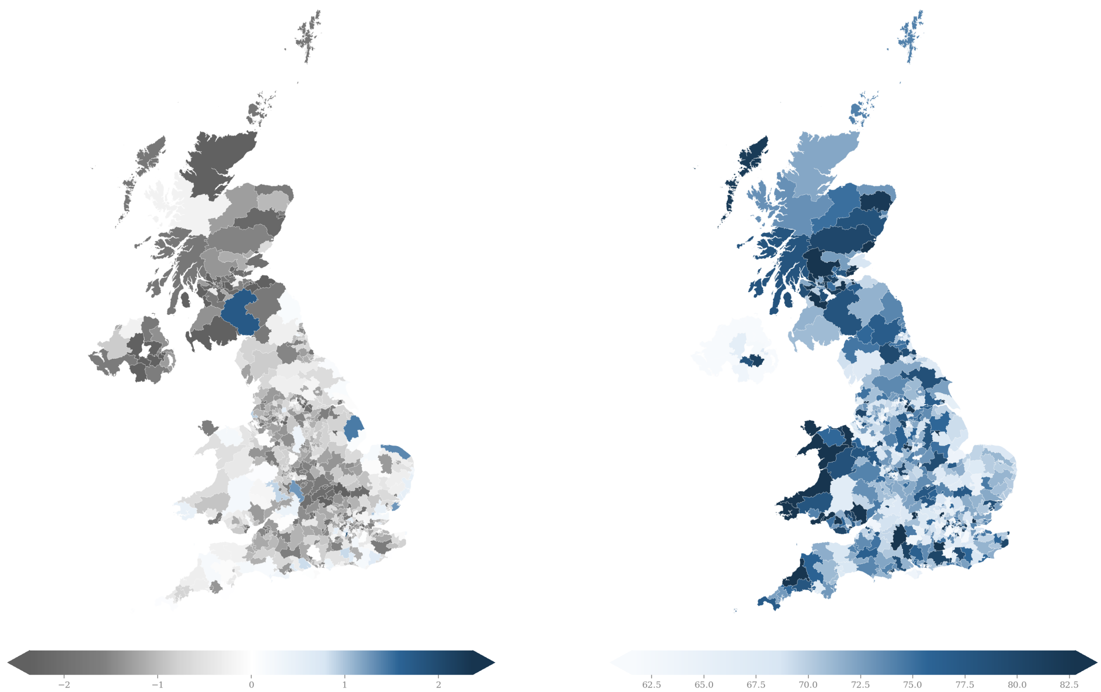

Figure 9: Constituency choropleth maps of the central scenario: change in aggregate household income (left) and displaced workers per 1,000 (right), enhanced FRS 2023–24 basis with imputed occupations, single draw (seed 0). Not directly comparable to the main-paper poverty numbers. The diverging scale is clipped at the 95th percentile of the absolute change (caps show clipped seats), so the English gradient is not crushed by a handful of extreme Northern Irish and inner-London constituencies. Because occupation is imputed within demographic cells rather than observed locally, most cross-constituency variation is within-region compositional scatter rather than a smooth spatial trend (see Appendix A.8).

4.7 The age and gender dimensions

The incidence axis has a within-family age structure worth drawing out. Under exposure-proportional incidence, workers aged 16–24 account for only 4.9 per cent of the displaced in the central scenario, because young people in the UK are concentrated in low-exposure service occupations; the burden falls on prime-age workers (28.1 per cent of the displaced are aged 35–44). The junior-concentrated family of Table 2 reweights displacement probabilities towards junior workers while holding the aggregate displacement rate at 7 per cent, raising the 16–24 share to 6.2 per cent and lowering the 55–64 share from 19.1 to 16.2 per cent. The aggregate consequences are statistically nil—over 20 draws the junior family is indistinguishable from the exposure-proportional one on cost (£18.3 ± 1.3 against £18.2 ± 1.5 billion), poverty and the Gini alike—but the exercise makes a structural point: seniority bias must be imposed, because it cannot be derived from occupational exposure alone. Even under the junior tilt, prime-age workers remain the largest displaced group.

If AI adoption in practice falls hardest on juniors through recruitment rather than dismissal—as the seniority-biased hiring evidence of Hosseini and Lichtinger (2026a) and the early-career employment declines in AI-exposed occupations documented by Brynjolfsson et al. (2025a) suggest—the fiscal consequences would be far smaller than the broad-based scenarios above, because junior workers contribute little income tax and National Insurance. That channel, however, operates through hiring flows that stock-based simulation frameworks such as mine do not capture by construction.

Exposure also carries a gender gradient. Employed women in the FRS have a mean C-AIOE of 0.33 against 0.18 for employed men, reflecting women's concentration in administrative, professional and associate-professional work. In the central scenario this translates into a displacement rate of 7.5 per cent for employed women against 6.4 per cent for men (means over 20 draws): women hold 51.8 per cent of employment but account for 55.6 per cent of the displaced. Household-level income changes are nevertheless similar by sex (−3.0 per cent for women, −2.8 per cent for men). A natural reading is that income pooling within couples mutes the individual asymmetry at the household level, though I have not decomposed the result and offer this as a hypothesis rather than a finding. The individual-level asymmetry itself echoes the international evidence that female employment is substantially more exposed to generative AI than male employment (ILO, 2025).

4.8 Benefit caseloads

The fiscal aggregates above can also be read on the delivery side, as caseloads. In the central scenario the shock draws 434,000 benefit units into Universal Credit while 58,000 exit, a net caseload increase of 376,000 benefit units, and annual UC spending rises by £3.3 billion from a baseline of £41.4 billion, of which £1.1 billion is the housing element within paid awards. Against DWP's roughly £334 billion of welfare spending in 2025–26 (a context figure, not a model output), the central increment is about 1 per cent; the high scenario roughly doubles it (695,000 net new benefit units, +£6.6 billion), while the low scenario adds only 58,000 benefit units and £0.4 billion. Caseload responses vary with incidence in the expected direction: the junior-concentrated family, whose displaced workers more often live in households with other earners, adds fewer benefit units (302,000 net) at lower cost (+£2.5 billion). Two accounting notes apply: only the benefit-unit figures are net of exits—the corresponding household and person counts (415,000 and 1.0 million in the central scenario) are gross new entrants—and the housing element is measured within paid awards, slightly overstating housing-specific cash.

5 Policy responses

If the shock modelled above is structural rather than cyclical—a permanent reallocation of labour demand rather than a downturn that unwinds—then the question facing the state is not whether the existing tax-benefit system cushions the blow (Section 4 shows that it does, absorbing roughly half of the market-income loss) but whether it cushions it well, and at what fiscal cost the residual hardship could be bought down. This section evaluates three stylised policy responses inside the shocked world. Each reform is simulated on top of the central scenario (7 per cent displacement, seed 0, 2026), so the estimand throughout is the difference between the shocked-plus-reform and shocked simulations: what the reform adds, given that the shock has happened. I evaluate them only as responses to the shock, not as standing policy; each is modelled for 2026 alone.

5.1 Three stylised reforms

The first reform (R1) is a wage-insurance scheme in the spirit of trade-adjustment proposals: displaced workers receive 50 per cent of their lost earnings, capped at £15,000 per year. I implement it as a post-simulation transfer that is non-taxable and disregarded by means tests, so its gross cost equals its net cost; this brackets the generous end of plausible implementations. The second (R2) is a demand-side circuit breaker within the existing architecture: a 20 per cent increase in the Universal Credit standard allowance, implemented as a parameter reform inside PolicyEngine so that all interactions with tapers, caps and passported entitlements are captured. The third (R3) suspends the benefit cap and cuts the UC earnings taper from 55 to 45 per cent, targeting the households for whom the existing system's own restraints bind hardest. R2 and R3 costs are net of clawback through the rest of the system.

Table 3: Policy reforms on top of the central shock, 2026 (exposure-proportional incidence, seed 0)

Reform	Annual cost (£bn)	Poverty change (pp) BHC	AHC	Gini change (pp)	Cost per pp averted (£bn) BHC	AHC
R1 Wage insurance	21.2	−1.53	−1.65	−0.97	13.8	12.8
R2 UC allowance +20%	5.8	−0.64	−0.56	−0.31	9.1	10.4
R3 Cap suspension + taper cut	4.7	−0.46	−0.45	−0.23	10.2	10.3

Note: All effects are shocked-plus-reform minus shocked, fixed seed-0 draw, reforms in force for 2026 only. R1: 50 per cent of lost earnings, £15,000 cap, paid to 1.61 million displaced workers (mean transfer £13,173), modelled as non-taxable and means-test-disregarded so gross cost equals net cost. R2 and R3 are PolicyEngine parameter reforms; costs are net of clawback. R3 combines benefit-cap suspension with a UC taper cut from 55 to 45 per cent. Poverty lines are held at their shocked-world values.

Table 3 reports the results and Figure 10 plots the decile pattern of the gains. The headline comparison is one of cost-effectiveness. The UC uplift averts a percentage point of after-housing-costs poverty for £10.4 billion, and the cap-and-taper package for £10.3 billion, against £12.8 billion for wage insurance. The reason is mechanical but instructive: wage insurance is earnings-proportional, and because the exposure-proportional shock displaces workers who are disproportionately in the upper half of the distribution, roughly 51 per cent of the R1 benefit is experienced in baseline deciles 8–10 (measured as a person-weighted household-gain share rather than a strict pound share). The UC-side instruments, by contrast, concentrate 71 per cent (R2) and 67 per cent (R3) of their benefit in deciles 1–5. Wage insurance does buy roughly three times more total poverty reduction than either UC reform—it is by far the largest intervention—but at roughly four times the cost, and with the weakest targeting per pound.

The reform comparisons are far less sensitive to the displacement draw than the shock aggregates: over 20 Monte Carlo draws the wage-insurance cost varies by only ±£0.2 billion (£21.4 $\pm$ 0.2 billion under exposure incidence, averting 1.44 ± 0.10 BHC points), and the UC-side instruments by less (R2: £5.69 $\pm$ 0.03 billion, -0.54 ± 0.02 points; R3: £4.57 $\pm$ 0.06 billion, -0.47 ± 0.03 points), so the cost-effectiveness ranking survives the draw noise comfortably. It also survives—indeed strengthens under—the six-month duration variant of Section 4.3: with displaced workers recovering half a year's earnings, a duration-consistent wage insurance costs

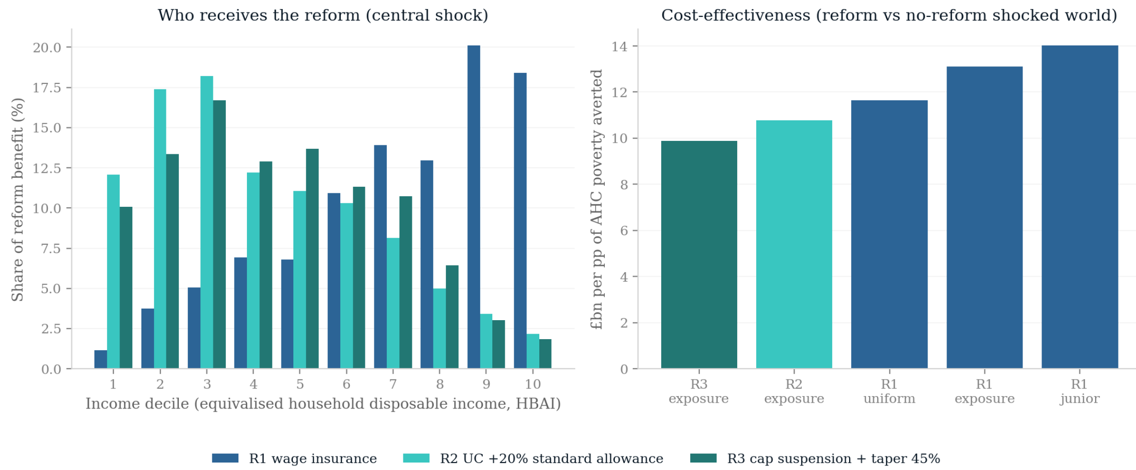

Figure 10: The three policy reforms on top of the central shock: fiscal cost, poverty and Gini effects, and the decile distribution of household gains. All panels show shocked-plus-reform minus shocked, seed 0, 2026.

£33 ± 6 billion per BHC poverty point averted against £11.0 ± 0.6 billion for the UC uplift, because shortening the earnings loss shrinks the poverty problem faster than it shrinks the earnings-proportional transfer. Because Section 4.5 established that incidence is the paper's central uncertainty, I re-run the wage-insurance reform under the junior-concentrated and uniform incidence families. The cost-effectiveness ranking is robust: R1 costs £14.3 billion per AHC point under junior incidence and £11.8 billion under uniform incidence, so even in its most favourable incidence world wage insurance only just reaches the £10.3–10.4 billion achieved by the UC-side instruments under the central family. The gross cost of R1 is nearly invariant across families (£20.3–21.4 billion), as it must be when the aggregate displaced wage bill is held approximately fixed; what varies is how much of that spending lands near the poverty line.

None of this settles the choice of instrument. Wage insurance addresses a different objective—consumption smoothing and earnings-loss compensation for the displaced themselves, most of whom are not poor—and a government persuaded by that objective would not be deterred by an unfavourable poverty-targeting ratio. What the exercise shows is narrower: if the criterion is poverty averted per pound in the shocked world, the existing means-tested architecture, temporarily made more generous, dominates a new earnings-linked scheme, precisely because the shock modelled here falls disproportionately on households that means tests are designed to exclude.

5.2 The tax-composition channel

The reforms above spend money into a fiscal position that the shock has already damaged, and the size of that damage depends on a question the microsimulation cannot answer by itself: where does the displaced wage bill go? The OBR's recent analysis of AI-related fiscal risks (OBR, 2026) emphasises that labour displacement erodes the income-tax and National Insurance base even if aggregate output is unchanged, with the fiscal outcome hinging on how much of the lost labour income reappears as taxable profit. I price this channel on top of the microsimulation with a simple accounting exercise. Let ϕ denote the share of the displaced wage bill that reappears as corporate profits taxed at the 25 per cent main rate. This exercise runs on the displacement-only, narrow-tax baseline: it prices the displaced wage bill and applies no wage uplift, so the £21.5 billion $\phi = 0$ shortfall below is the 7 per cent/no-uplift cash-tax cell of Section 4.4, not the £18.2 billion full-balance central. On that baseline the displaced wage bill is £77.9 billion and the effective income-tax-plus-NICs rate on displaced workers' earnings is 25.0 per cent, so the labour tax forgone is £19.5 billion—and because that effective labour rate almost exactly equals the corporation-tax main rate, the shortfall is very nearly $(1 - \phi) \times 0.25\times$ the displaced wage bill. At $\phi = 0$ the net revenue shortfall is the microsimulation's £21.5 billion; at $\phi = 0.5$ it falls to £11.8 billion; and at $\phi = 1$ the labour-to-capital channel is almost exactly self-financing (a £0.03 billion surplus relative to full labour taxation), leaving only a residual £2.1 billion net cost, chiefly shock-induced benefit spending together with smaller non-labour-tax revenue changes. Figure 11 traces the full grid over displacement rates and ϕ.

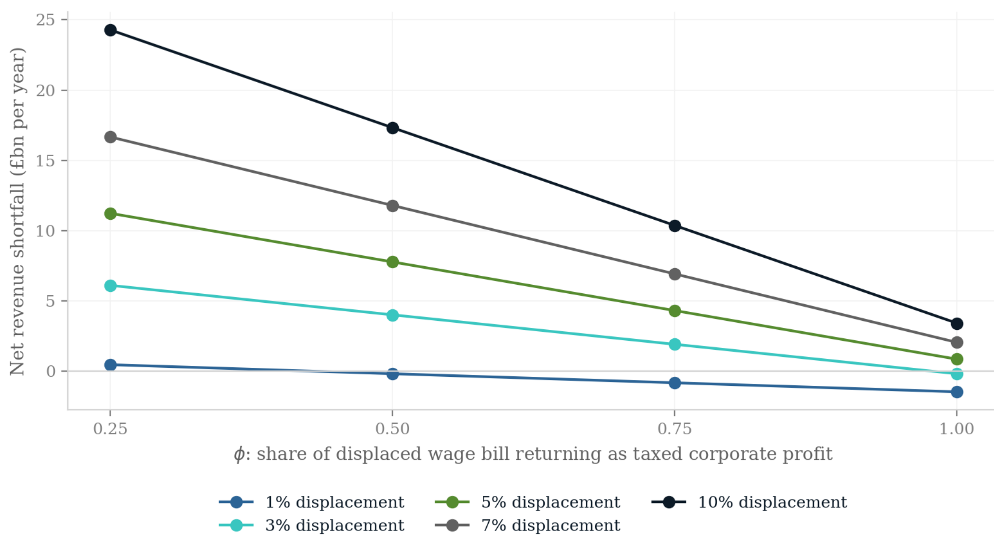

Figure 11: Net revenue change by displacement rate and ϕ, the share of the displaced wage bill reappearing as corporate profits taxed at 25 per cent. The $\phi = 0$ line is the microsimulation's net revenue change; higher ϕ adds recouped corporation tax outside the simulation.

Revenue neutrality, however, is not distributional neutrality. To see who receives the recomposed income, I simulate a dividend-recycling case at $\phi = 0.5$ with a payout ratio of 0.5 (a parameter, not an estimate): £14.6 billion of post-CT profits are distributed pro rata to existing FRS dividend holders, with the dividend tax on those distributions captured inside the simulation. The result is stark. The Gini rises by a further 0.62 percentage points relative to the shocked world without recycling, and poverty—before or after housing costs—moves by exactly zero: the recycled dividends reach no household near the poverty line. The decile pattern of the gains is lumpy because FRS dividend records are sparse, but the message is not: deciles 1–4 receive exactly nothing. Even in the most fiscally benign reading of the shock, in which the Exchequer recovers essentially all of the lost labour tax through the corporate channel, the compositional shift from wages to capital income leaves the bottom of the distribution no better off and the overall distribution more unequal—which is precisely the configuration in which the transfer-side instruments of Section 5.1 would be needed most.

Two caveats bound this exercise. It is static accounting layered on top of the microsimulation: the incidence of corporation tax is not modelled, the corporation-tax base is assumed equal to accounting profit (no allowances, losses or profit shifting), and the effective labour rate is a pro-rata attribution of income tax and NICs to employment income that ignores the schedule ordering of savings and dividend bands. And ϕ itself is the free parameter the whole channel turns on; I present a grid rather than a point estimate because nothing in my framework pins it down. Relatedly, the $\phi = 1$ result is closer to an accounting identity than an empirical finding; the object of interest is the distributional consequence at every ϕ, not the revenue arithmetic at the endpoint.

6 Conclusions and policy implications

The results should be read as scenario analysis, not prediction. I do not forecast the pace or extent of AI adoption in the United Kingdom; I trace the distributional and fiscal consequences that would follow *if* labour-market shocks of a given size and occupational incidence were to materialise, under tax-benefit rules as legislated. The observational record remains too thin to adjudicate between the incidence families: the earliest evidence ranges from minimal measured earnings and hours effects to date at one extreme to a modest relative contraction in exposed occupations at the other, none yet at the scale of my central shock. That unsettled state is the case for spanning the disagreement rather than resolving it by assumption, and what the paper contributes is not a claim about which world will arrive but a quantification of what turns on the answer.

Two structural features, established in Sections 4.3–4.5, give the results their shape and survive everything the paper varies. The tax-benefit system supplies substantial automatic insurance—means-tested support absorbs market-income losses low in the distribution, progressive taxation forgoes revenue higher up—but that protection is purchased through a deterioration of the public finances rather than prevented: the losses are redistributed, not averted. And because the Gini re-ranks households onto post-shock incomes, measured inequality rises in every shock-size cell and every incidence family even where the market shock falls hardest on the top deciles (Figure 1); the same re-ranking, documented for Ireland by Doorley et al. (2026), is what sets an AI shock aimed at cognitive work apart from the routine-biased automation of earlier waves (Autor et al., 2003; Doorley et al., 2023). The consequence is that neither the fiscal cost nor the rise in inequality is an artefact of a particular scenario; both are properties of how a labour-taxation-heavy welfare state absorbs a top-loaded shock.

6.1 Policy implications: what depends on who bears the shock

The central policy message follows from the incidence axis rather than from any single scenario. Inequality rises whichever family one assumes; what the incidence determines is whether the UK faces primarily a fiscal-resilience problem or a poverty-protection problem.

If the shock is top-loaded, as in the exposure-proportional family, the problem is chiefly fiscal. Displaced workers are drawn from occupations that pay well and are taxed heavily, so each displacement removes a large slug of income tax and National Insurance while adding comparatively little to means-tested expenditure. The Exchequer cost of the central shock is £18.2 billion under exposure-proportional incidence against £14.9 billion under uniform incidence (20-draw means)—the same aggregate shock, over a fifth more expensive purely because of who bears it. Under top-loaded incidence the first-order policy question is revenue resilience: the UK tax base leans heavily on the taxation of labour income, and the workers most exposed are precisely those who contribute most to it. If AI additionally shifts the functional distribution of income from labour towards capital, the base that finances the cushioning I document would itself erode—a possibility that has motivated recent work on the taxation of capital in an AI-intensive economy (Bastani and Waldenström, 2024). The tax-composition exercise of Section 5 prices that possibility for the Treasury: whether base erosion is a first-order fiscal problem turns entirely on how much of the displaced wage bill reappears as taxable corporate profit.

If instead the shock lands flatter or on junior workers, the fiscal exposure moderates but the poverty exposure does not. Uniform incidence is the cheapest family I run, yet its poverty effect (+1.84 points before housing costs) is statistically indistinguishable from the exposure family's (+1.75). Junior-concentrated incidence is indistinguishable from the exposure family on cost and poverty alike. In these worlds the binding constraint is the adequacy of out-of-work support, and the relevant instruments are benefit levels, the contributory window, and—for the junior case—the entry prospects of young workers. Evidence that generative AI operates as a seniority-biased technology, compressing demand for junior and entry-level roles in exposed occupations (Hosseini and Lichtinger, 2026a; Brynjolfsson et al., 2025a), with early UK corroboration in Klein Teeselink (2025), argues for close monitoring of youth labour-market entry. My junior-concentrated family can only re-weight who is displaced within a single year; the dynamic costs of a scarred entry cohort—foregone experience, delayed progression, persistent earnings losses—lie outside a static framework, so even that family understates the long-run burden on new entrants.

The fiscal-versus-poverty framing has a policy answer, not only a diagnosis. The counterfactual reforms of Section 5 show that temporarily enlarging the existing means-tested architecture—a circuit-breaker uplift to the Universal Credit standard allowance, or a benefit-cap suspension combined with a taper cut—averts poverty more cheaply per pound than a new earnings-linked wage-insurance scheme, precisely because the shock falls disproportionately on households that means tests are designed to exclude. The implication for a government facing a top-loaded shock and a strained revenue base is to reach first for the stabilisers it already operates rather than build an earnings-replacement scheme alongside them.

The expertise-compression family, standing in for the view that AI erodes the returns to accumulated expertise, is the worst of both worlds among the stylised families: clearly the most expensive (£22.1 billion, an ordering that survives the draw noise) and, on the point estimate, the most poverty-raising (+1.93 points, though the poverty ordering among the stylised families does not survive one standard deviation)—and the measured family exceeds it on both margins (Table 2). If that view proves correct, neither fiscal buffers nor poverty protection alone suffices; both margins bind at once. This household-level result is not inconsistent with Hosseini and Lichtinger (2026b), who find that the pure labour-supply channel is equalising for between-occupation wages: the Gini rise here reflects the displacement-based implementation and the capital channel, not a claim that expertise compression widens occupational wage dispersion.

The shock also has a geography, and it does not track existing regional-policy maps. In the constituency analysis of Section 4.6, the largest proportionate income losses fall on Northern Ireland—the worst-hit region, at −2.3 per cent of aggregate household income—and Scotland, while Wales and the South East are mildest; the hardest-hit constituencies mix Northern Irish seats with inner-London ones. An AI labour shock is not a rerun of deindustrialisation: because exposure follows clerical and professional employment, it strikes administrative and professional centres wherever they sit, and a levelling-up strategy calibrated to the geography of manufacturing decline would miss much of it. These geographic results rest on imputed occupations on the enhanced 2023–24 dataset (Appendix A.8) and are indicative of relative, not absolute, local impacts.

Two implications hold across all families. First, because the modelled losses fall on identifiable occupational groups rather than diffusely across the workforce, retraining and lifelong-learning provision targeted at exposed occupations is the most direct lever available—and the earliest US evidence suggests it can work: Hyman et al. (2025) find that earnings returns to publicly funded retraining for workers from highly AI-exposed occupations roughly tripled during the 2022–2024 AI boom, largely through transitions into less-exposed work, and estimate that 40–45 per cent of occupations are 'AI-retrainable'. The tax-benefit system can replace income but cannot restore occupational attachment. Second, since measured inequality rises under every incidence I examine, distributional monitoring should not wait for evidence on which incidence is materialising: the direction of the inequality effect is not in doubt within this framework, only its composition.

Taken together, the tax-benefit system converts each market-income shock into a smaller disposable-income shock at a substantial, incidence-dependent fiscal cost, while the polarising character of displacement pushes measured inequality upwards regardless of who bears it. Policy attention is best directed at the margins the system does not reach: the occupational reallocation of exposed workers, the entry prospects of the young, and the robustness of a labour-taxation-heavy revenue base.

6.2 Limitations and further work

Several limitations qualify the results; I state them by pipeline stage, with the measurement conventions needed to compare my figures with official statistics.

The first stage is the data and its weighting. All results use the HBAI equivalised household disposable income concept, with the Gini computed after bottom-coding negative incomes at zero. My baseline Gini of 0.309 sits below the official 0.34 because I use the plain Family Resources Survey without the Survey of Personal Incomes (SPI) top-income adjustment, so levels are not comparable to official statistics even though the changes I report are computed on a consistent basis. I use the uncalibrated FRS grossing weights rather than PolicyEngine's enhanced dataset, whose household cloning would break the record-to-occupation correspondence the exposure join depends on; aggregate fiscal magnitudes therefore carry the usual survey-grossing caveats. Single-draw figures also carry displacement-draw noise that the 20-draw averages remove (seed conventions in Section 3.5 and Appendix A.1).

Occupational exposure is assigned at the 1-digit SOC level, because the FRS records occupation only at the major-group level, so within-group heterogeneity in exposure (Felten et al., 2021; Webb, 2019; Eloundou et al., 2024) is averaged away. The scores also reach the FRS through a reconstructed pipeline—a rebuilt complementarity adjustment and a chained SOC2018–SOC2010–ISCO–SOC2020 crosswalk with ASHE weights for 340 of 412 unit groups. Adopting the UK-native GAISI index (Henseke et al., 2025) together with detailed-occupation imputation from the Labour Force Survey would remove both the crosswalk and the resolution constraint, and is my first priority; for the expertise-compression family specifically, the public occupation-level skill-specificity and expertise measures of Hosseini and Lichtinger (2026b) offer a further planned robustness input.

The displacement treatment is static: displaced workers are fully out of work for the simulated year (earnings, hours, pension contributions and salary sacrifice zeroed, entitlements recomputed), while dual jobholders keep self-employment income. Unemployment duration is not modelled, so contributory-benefit exhaustion (new-style Jobseeker's Allowance lasts at most six months) is not captured, and over longer horizons I overstate support for displaced workers without means-tested entitlements. The sensitivity runs of Section 4.3 show duration is first-order—a six-month spell cuts the fiscal cost by more than three-quarters—whereas incomplete take-up barely moves the headline. Nor is there a wage-adjustment margin for displaced workers: in equilibrium task models (Acemoglu and Restrepo, 2022), displacement lowers relative wages and triggers ripple effects onto non-displaced workers rather than producing spells of zero earnings, so my framework overstates the unemployment channel and understates within-employment wage dispersion.

The capital channel is narrow: it covers only interest and dividends, excluding rental income and returns inside private pensions, and surveys under-represent top and capital incomes, so the top-decile concentration that drives this channel is if anything understated. Linked wealth data (the Wealth and Assets Survey) would sharpen it.

All scenarios exclude the self-employed, over whom the exposure indices and displacement literature are not defined; since self-employment is a meaningful, partly exposed share of UK employment, this is a substantive omission.

Finally, the mechanisms I invoke for decile-level patterns—thin savings buffers at the bottom, income pooling within couples—are hypotheses consistent with the data, not formal decompositions, and the scenario anchors are calibration conventions: the 1 per cent "low" cell is a sensitivity case, and the 7 per cent displacement and 2.6 per cent wage figures convert task-exposure and productivity estimates into single-year shocks. As UK-specific evidence accumulates (Klein Teeselink, 2025; Rockall et al., 2025) the scenarios can be re-anchored. But the central lesson does not wait on it: on whichever occupations an AI displacement shock ultimately falls, a welfare state financed by taxes on labour income can redistribute the resulting losses effectively but cannot, and will not, prevent them.

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A Additional tables and figures

A.1 Seed and draw conventions

Displacement is a random draw of which exposed workers lose employment, subject to occupation-group quotas proportional to employment times mean exposure. Throughout the paper, three conventions apply. The scenario grid (Figures 5–7, and their appendix variants below), the decomposition in Figure 4 and the preset scenarios are single draws at seed 0, while the five incidence families report means over 20 draws (seeds 0–19). The decile transition and wage-gain profiles (Figures 1 and 2) are means over 50 independent draws. The robustness summary and the confidence intervals in Figure 19 are 20-draw Monte Carlo runs, with ± values reported as standard deviations across draws. Within each draw, ordering keys are uniform within occupation group, so a record's inclusion probability does not depend on its grossing weight.

A.2 Job loss by occupation group

Table 4 reports the central-scenario displacement rate for each SOC2020 major group. The allocation rule concentrates losses in administrative and secretarial work (10.5 per cent of the group's employment), managerial and professional occupations (around 9.5 per cent) and associate professional roles, while elementary occupations—the least-exposed group, whose normalised exposure score is zero—experience no displacement at all. Clerical and professional work bears the shock; manual and elementary work is insulated.

Table 4: Central-scenario job loss by SOC2020 major group

Major group	C-AIOE	Employment (m)	Job loss (%)
1 Managers, directors and senior officials	0.62	2.45	9.6
2 Professional occupations	0.60	6.46	9.5
3 Associate professional occupations	0.47	3.13	8.5
4 Administrative and secretarial occupations	0.74	2.35	10.5
5 Skilled trades occupations	−0.33	1.56	2.8
6 Caring, leisure and other service occupations	−0.14	2.32	3.9
7 Sales and customer service occupations	0.27	1.22	7.2
8 Process, plant and machine operatives	−0.54	1.41	1.3
9 Elementary occupations	−0.73	2.05	0.0

C-AIOE is the standardised complementarity-adjusted exposure score before the min-zero normalisation used in the allocation rule. Employment and job-loss rates are survey-weighted; single draw, seed 0.

A.3 Where the shock lands: baseline distributions

Figures 12–16 show the baseline household-level decile distributions of the income components through which the three channels operate (aggregate £billion per year by decile of equivalised household disposable income). Market income excluding capital (£1,313 billion in total) and income tax and National Insurance (£312 billion) are heavily concentrated in the top deciles—which is why displacement of high earners is fiscally expensive. Benefits (£320 billion) are spread across the lower and middle deciles, peaking in deciles 3 to 5. Capital income (£23 billion of interest and dividends) is the most concentrated series of all: the top decile holds more than a third of the total.

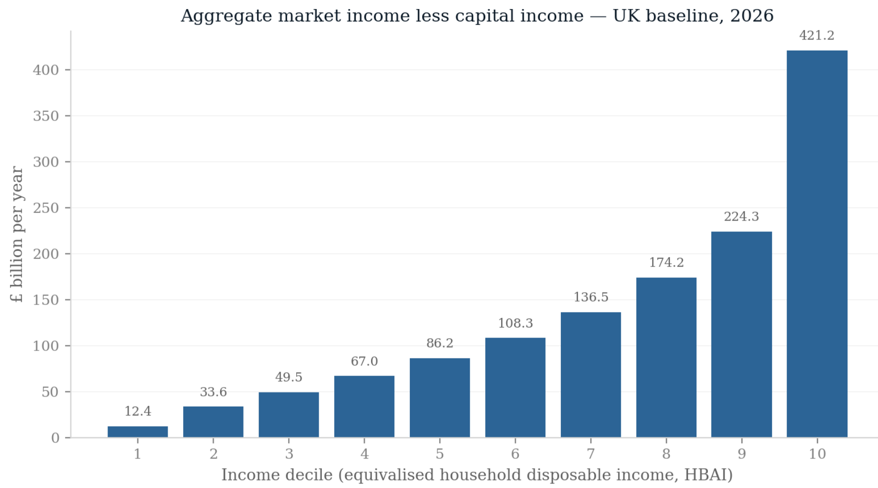

Figure 12: Baseline aggregate household market income (less capital income) by decile.

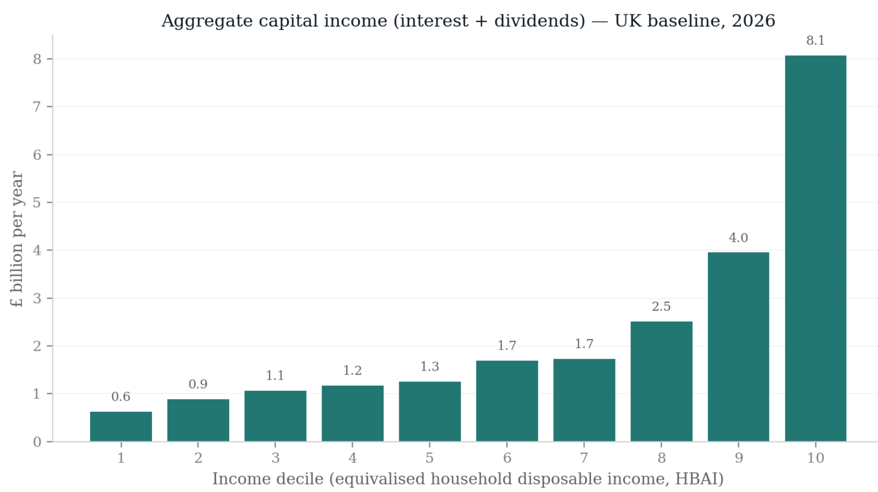

Figure 13: Baseline aggregate household capital income (interest and dividends) by decile.

A.4 Exposure incidence before any simulation

Figures 17 and 18 show, without any shock, how workers in each income decile distribute across employment-weighted tertiles of AI exposure and of complementarity. The share of workers in the top exposure tertile rises steadily with household income, while complementarity shows no comparable gradient—the descriptive facts that generate the simulated incidence in Section 4.

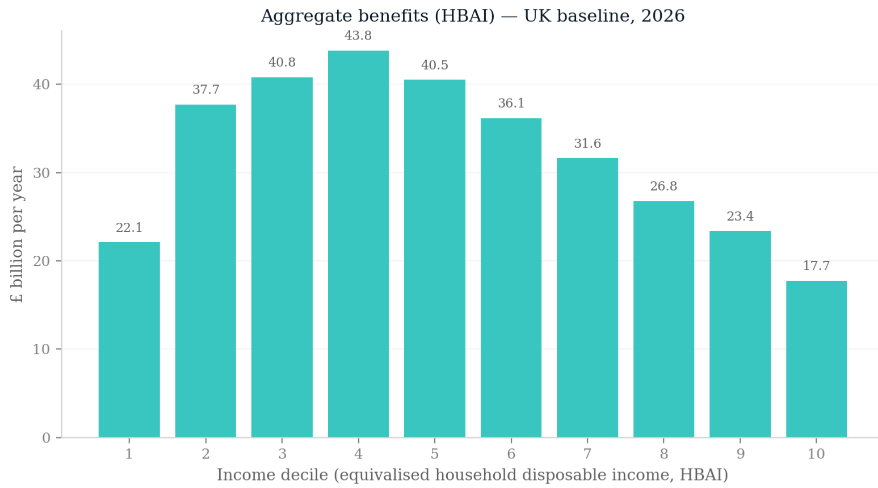

Figure 14: Baseline aggregate household benefits by decile.

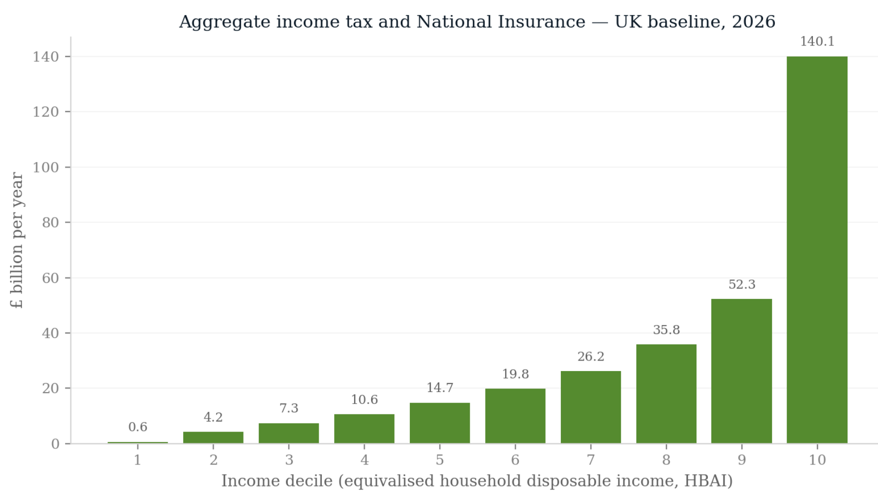

Figure 15: Baseline aggregate household income tax and National Insurance by decile.

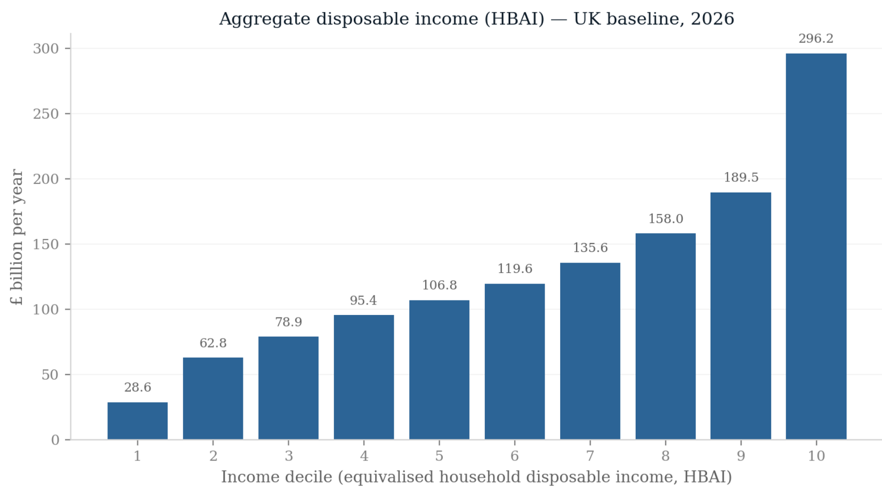

Figure 16: Baseline aggregate household disposable income (HBAI concept) by decile.

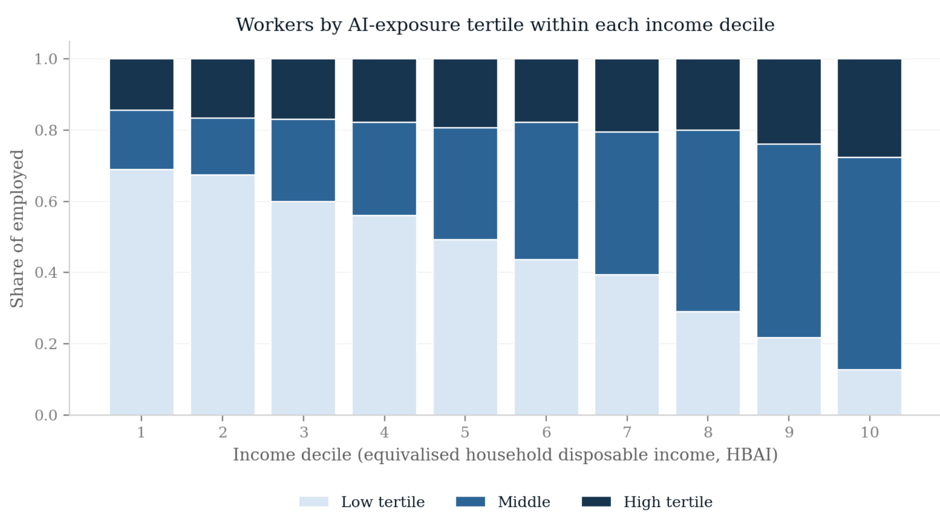

Figure 17: Distribution of employed persons across AI-exposure tertiles, by income decile.

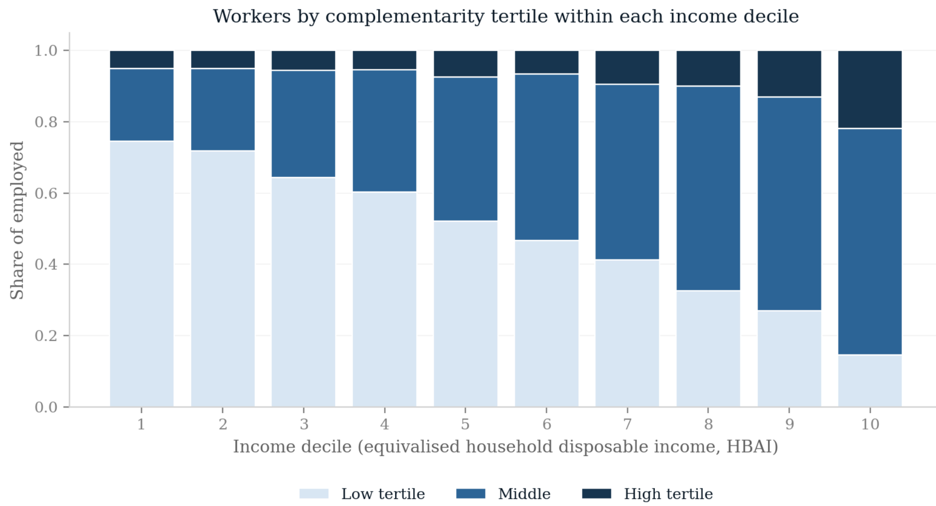

Figure 18: Distribution of employed persons across complementarity tertiles, by income decile.

A.5 Monte Carlo uncertainty on the decile profile

Figure 19 reports the central-scenario disposable-income change by decile as the mean over twenty independent displacement draws with 95 per cent confidence intervals. The gradient runs from −0.5 per cent in the bottom decile to −4.1 per cent in the top. It is not quite monotone: deciles 2 and 3 show a small reversal (−0.88 against −0.82 per cent), well inside the draw noise. The overall upward slope of the loss profile survives the uncertainty comfortably.

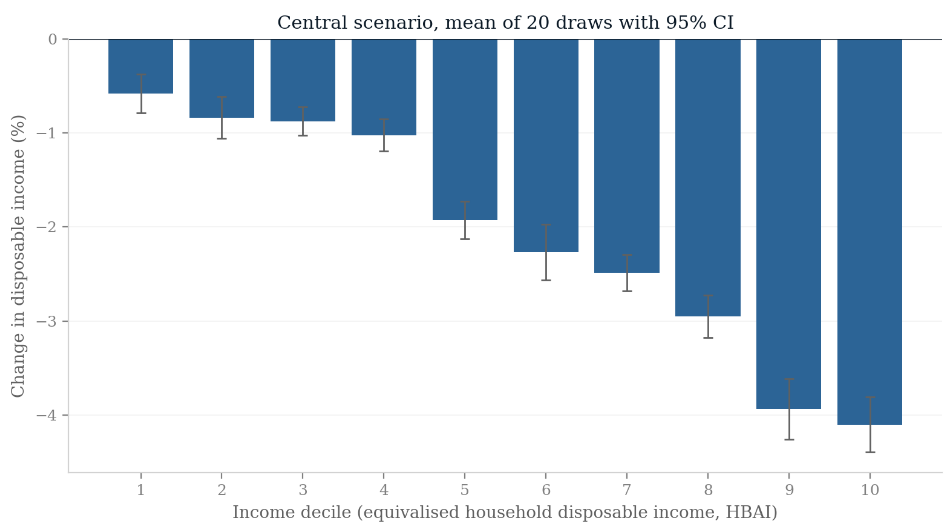

Figure 19: Disposable income change by decile, central scenario: mean of 20 displacement draws with 95 per cent confidence intervals.

A.6 Uniform-shock and alternative-index comparisons by decile

Figure 20 compares the central scenario against the same aggregate shocks allocated uniformly across occupations, decile by decile; Figure 21 repeats the central scenario with the Eloundou et al. (2024) GPT-exposure index in place of the C-AIOE. The exposure-based allocation shifts losses up the distribution relative to a uniform shock (top-decile losses of 3.9 against 3.0 per cent; bottom-decile losses of 0.5 against 0.9 per cent), and the choice of exposure index leaves the decile profile essentially unchanged.

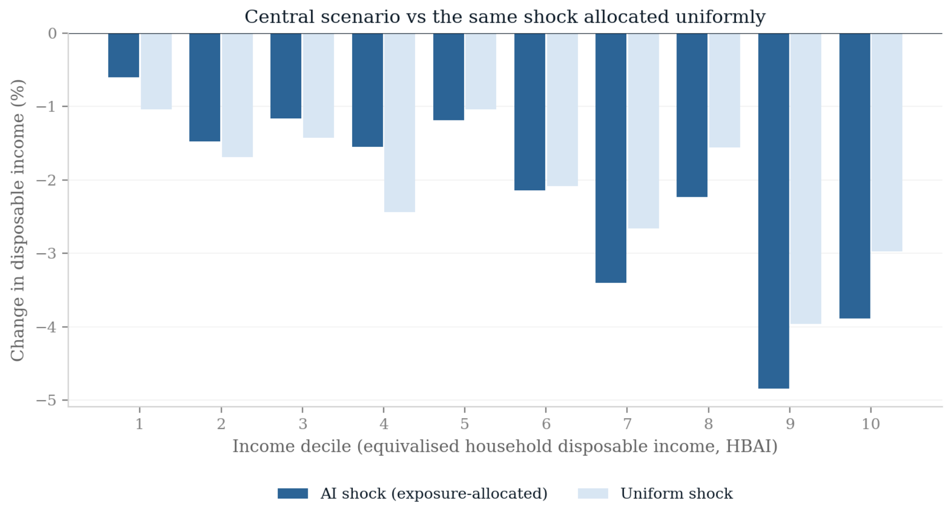

Figure 20: Disposable income change by decile: exposure-allocated versus uniform shock, central parameters.

A.7 Full exposure-index sensitivity

Figure 21 compared decile profiles under two indices; here I rerun the entire central scenario under five: the C-AIOE (my central measure), the unadjusted Felten et al. (2021) AIOE, the Eloundou et al. (2024) GPT-exposure beta, and the two indices from the UK government's DSIT exposure assessment (all-AI and large-language-model variants). The comparison matters because the published literature disagrees sharply on *levels*: estimates of the share of employment "highly exposed" to AI range from a few per cent to well over half depending on the index and threshold chosen, so any result that survived under only one index would be fragile. My quota mechanism holds the aggregate displacement rate at 7 per cent, so the indices differ only in *where* the same aggregate shock lands across occupation groups.

Table 5 and Figure 22 report the outcomes. The exchequer cost spans £17.6–18.6 billion (the largest deviation from the C-AIOE figure is −4.3 per cent, under the DSIT LLM index), the Gini change spans +1.05 to +1.10 points, and the before-housing-costs poverty change spans +1.85 to +1.90 points. The decile-10 transition share exceeds the decile-1 share by a factor of 8.9 to 11.3 across indices. All three qualitative conclusions of the paper—the Gini rises, the displacement-transition gradient rises with income, and the fiscal cost is within 20 per cent of the central estimate—hold under every index, essentially because all five rank SOC major groups

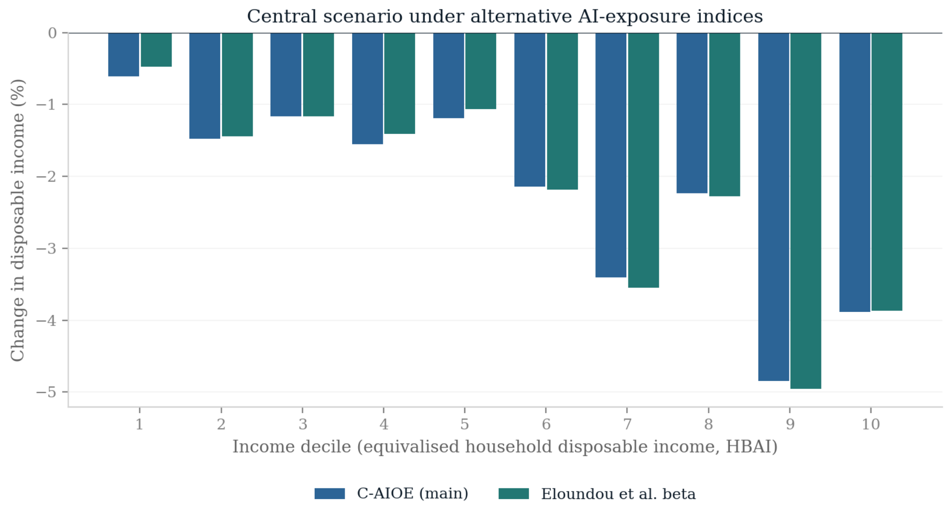

Figure 21: Disposable income change by decile under the C-AIOE and the Eloundou et al. exposure index.

similarly: clerical and professional work is most exposed under each, and the quota fixes the aggregate.

Table 5: Central scenario under five exposure indices

Index	Cost (£bn)	ΔPov. BHC (pp)	ΔGini (pp)	D1 share (%)	D10 share (%)
C-AIOE (central)	18.4	1.87	1.10	0.53	4.67
Felten AIOE	18.6	1.86	1.09	0.53	4.70
Eloundou beta	18.4	1.90	1.09	0.47	4.74
DSIT AIOE	18.6	1.86	1.05	0.43	4.86
DSIT LLM	17.6	1.85	1.09	0.51	4.68

Single draw, seed 0, 2026. D1 and D10 shares are the percentages of persons in the bottom and top baseline income deciles whose household is pushed down at least one decile by displacement. Aggregate displacement is held at 7 per cent by the quota mechanism under every index.

A.8 Data appendix: constituency-level occupation imputation

The constituency results use PolicyEngine's 650-constituency weight matrix, which indexes the enhanced FRS 2023–24 dataset (53,566 employee records); on that dataset the serial-number join to observed SOC major group is invalid, so occupation had to be imputed. I drew each enhanced record's SOC major group from the survey-weighted occupation distribution of the plain FRS 2024–25 (12,800 employees, 99.6 per cent with observed SOC) within cells defined by age band × gender × region × individual-earnings decile, at seed 0. On a weighted basis 99.4 per cent of employees were imputed from the full four-way cell, 0.6 per cent from an age–gender–decile fallback and 0.03 per cent from the marginal distribution. As stressed in the main text, occupation is therefore imputed rather than observed at the constituency level: cross-constituency variation reflects demographic and earnings composition (plus one seed-0 draw), not directly measured local occupation mixes, and the enhanced-dataset results are not directly

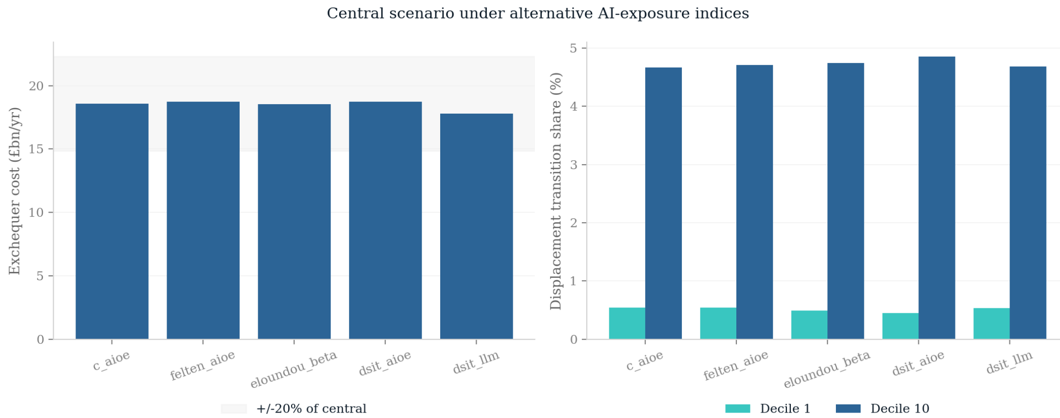

Figure 22: Headline outcomes of the central scenario under five exposure indices.

comparable to the plain-FRS poverty numbers elsewhere in the paper.

Why the poverty response roughly doubles. On the enhanced FRS 2023–24 dataset the national BHC poverty response to the central scenario is +3.9 percentage points, roughly twice the +1.87 points of the plain-FRS central run. Running the central scenario on the enhanced dataset at the main analysis's period attributes about +0.7 points of the gap to the dataset itself and +0.3 points to the simulation period, with the remainder arising in the constituency-weight aggregation.

Constituency ranking. The worst-hit constituencies are Lagan Valley (−5.43 per cent), Upper Bann (−4.11 per cent) and Belfast East (−4.03 per cent), with Stratford and Bow and Islington South and Finsbury close behind—a pattern combining Northern Irish seats with high-exposure inner-London ones. A small number of constituencies gain on net, led by Kingston and Surbiton (+2.05 per cent) and Dumfriesshire, Clydesdale and Tweeddale (+1.75 per cent), where wage and capital gains outweigh displacement losses.

A.9 The scenario grid by decile, and without the capital shock

Figure 23 facets the full 66-cell scenario grid by baseline income decile; cells report the change in mean household disposable income (HBAI concept), which across the grid as a whole ranges from +3.0 to −5.2 per cent. Every decile's panel shifts towards losses as displacement rises, but the top deciles darken much faster: in the harshest cell (10 per cent displacement, no wage gain) the bottom decile loses 1.9 per cent of disposable income while the top decile loses 8.2 per cent. Figure 24 removes the capital-return shock. Average disposable incomes fall relative to the capital-on grid, confirming that the capital channel raises average incomes while concentrating its gains at the top; the Gini result, however, is unchanged in kind: without the capital shock the Gini of equivalised disposable income still rises in all 66 cells, by 0.00 to 1.50 percentage points. The inequality result does not depend on the capital channel.

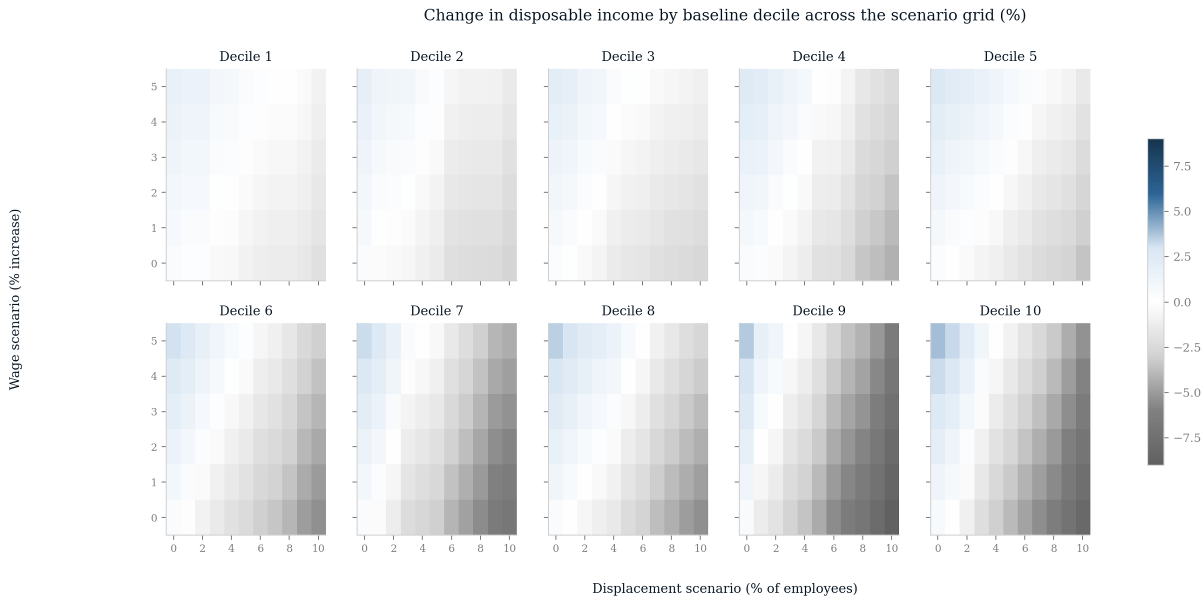

Figure 23: Change in household disposable income across the scenario grid, faceted by baseline income decile. Single draw, seed 0.

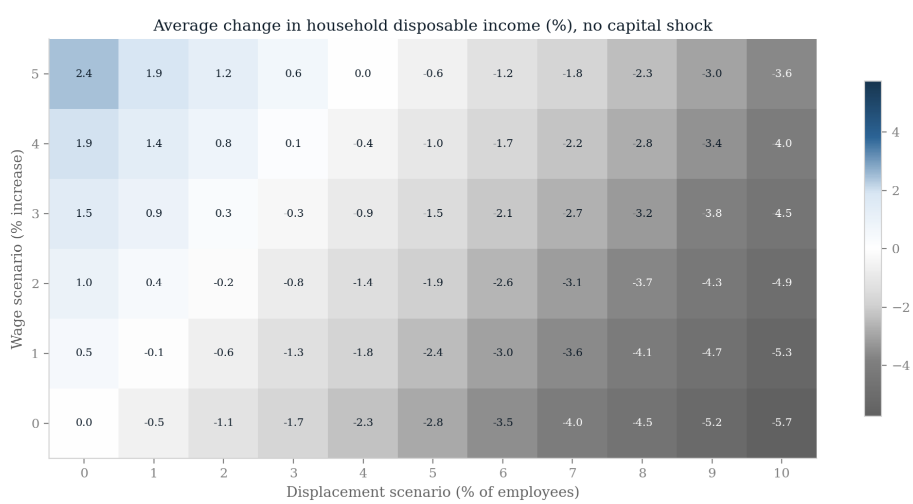

Figure 24: Average change in household disposable income across the scenario grid with the capital-return shock switched off. Single draw, seed 0.